Radiology
Workbook

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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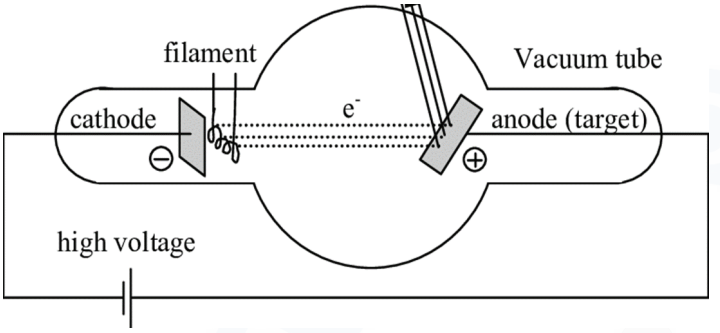
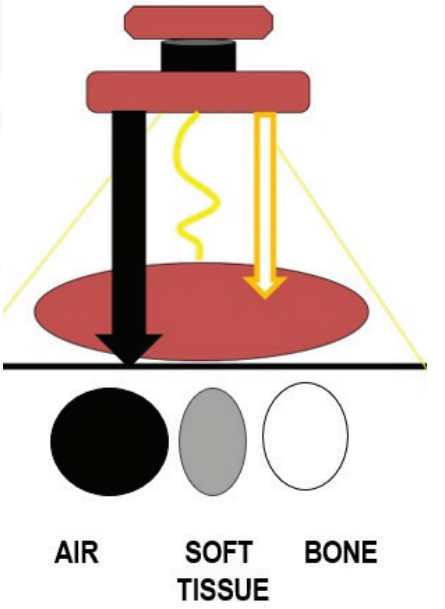
GENERAL RADIOLOGY

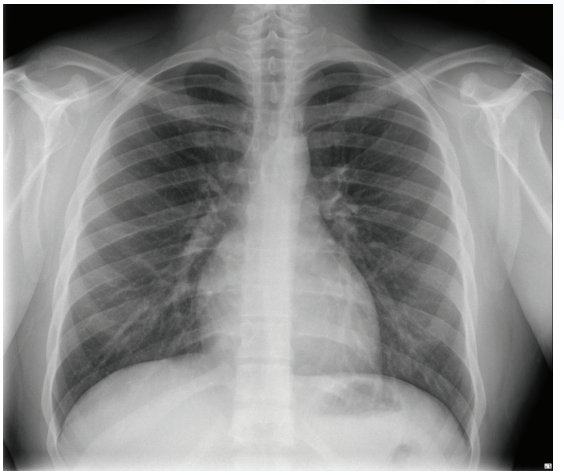
Radiography with Normal Chest X-ray





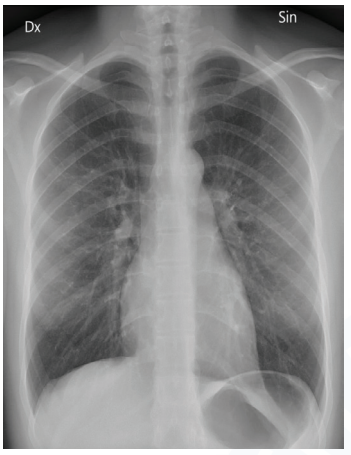
1. X-rays are produced by the-
2. Parts of X-ray Tube-
3. Cathode and Anode is made of –
4. Cathode produces -
5. Anode produces –

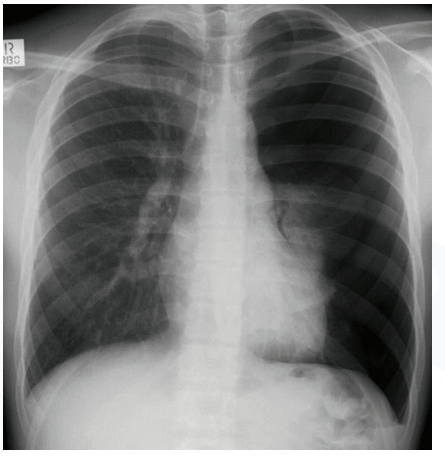
High energy X-Ray Tube is used in Radiotherapy Q- k/a



- Air density- Air anywhere will appear
- Bone density-
  - Anatomical location of bone
  - Non anatomical:
    Stone/Calculus
    Calcification
- Soft tissue- Anything which is not air density or bone density.

Radiographic density is normal for the location or abnormal..



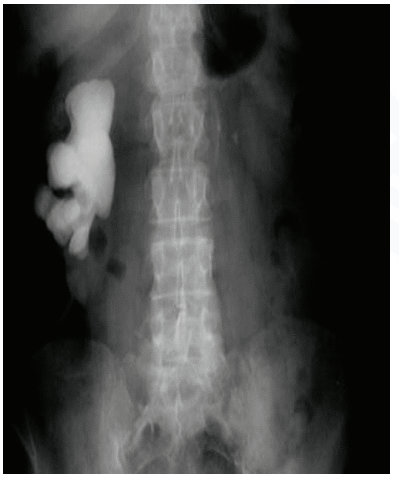


Lucency devoid of vascular markings on a CXR Indicates Q

Chest X-ray done in a patient with sudden Onset Abdominal Pain. Q

Always rule out ....

Non-anatomical Location of Bone density



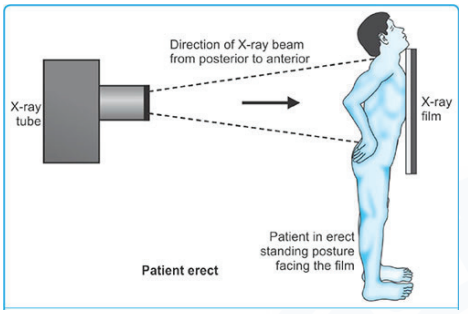
Bone density taking the shape of Lumen is

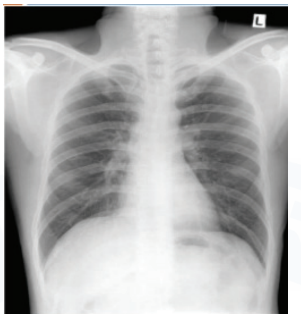
Bone density taking the shape of parenchyma is

Principle application - Radiography / X-ray

- **Densities** : 3 densities-Air / Bone / Soft tissue density
  We see the soft tissues but we are not able to differentiate different types of soft tissue
- **Terminologies**:Abnormal area of soft tissue density-
  Abnormal area of air density -
- **Initial investigation:** Air anywhere / Bone and Joint pathology /Foreign body

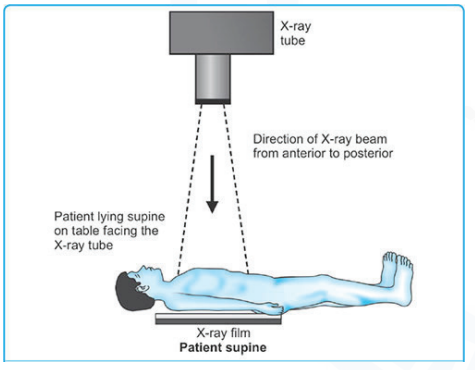
Normal Chest Radiograph:





CXR-PA View-

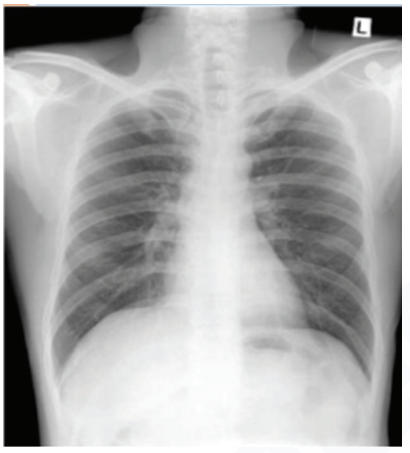
1. Direction of X-ray is from
2. Taken in position
3. Identify by-



CXR-AP view

1. Direction of the X-ray is from
2. Taken in Position
3. Indicated when patient Cannot stand

CXR-PA Vs. AP view



Position of the Clavicle

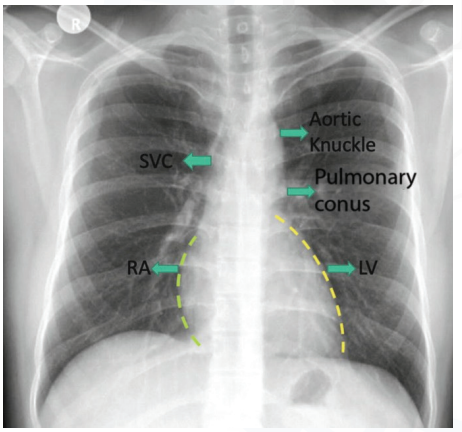
Oblique Horizontal

Scapula in relation to Lung Field

Not Overlapping vs Overlap

Fundal Gas Shadow

Cardiac Border



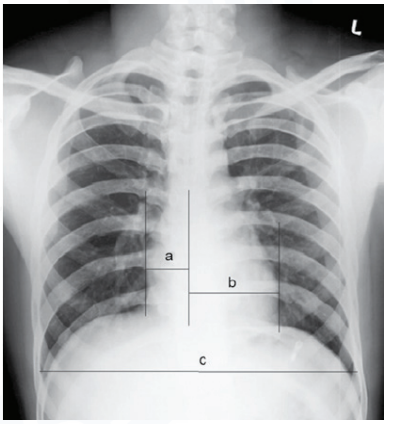
Right Heart Border formed by –

Which structure is NOT the part of Rt Heart Border Q

Left Heart Border formed by -

Which structure is NOT the part of Left Heart Border Q

Cardio-thoracic ratio (CTR)



Cardio-thoracic ratio- It is the ratio of Max. heart diameter To Max. internal thoracic diameter measured on CXR- PA view

$$\frac{a \text{ (Rt Heart Diameter)} + b \text{ (Left Heart Diameter)}}{C \text{ (Maximum Internal Thoracic Diameter)}}$$

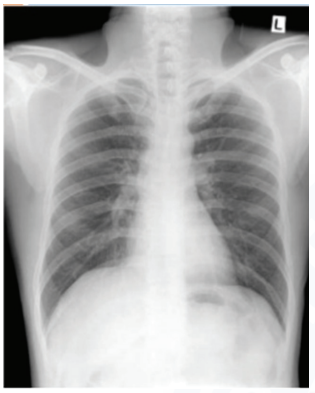
Cardiomegaly is present when CTR is more than on CXR View Q

Cardiac Apex and Base[Q]

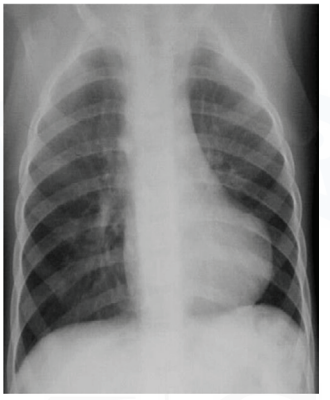
Apex- Outermost point on the left heart border- Formed By

Base formed by-

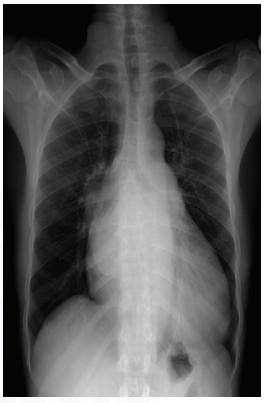
Normal LV Apex



Abnormal Apex



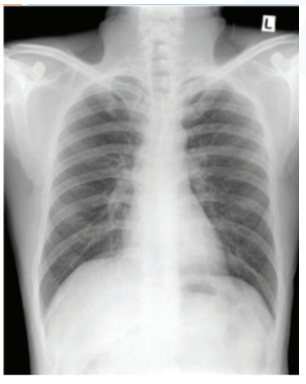
Abnormal Apex



Pulmonary Hila-

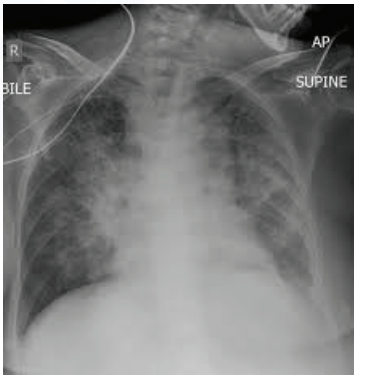
Lateral margin of Normal Pulmonary Hila is Concave

Normal Hila



Abnormal Hila

Opacity which can be confused with Abnormal Hila

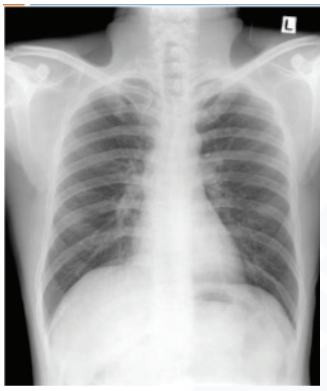


M/C cause of Bilateral Hilar Lymphadenopathy- [Q]

Costo-phrenic Angle:

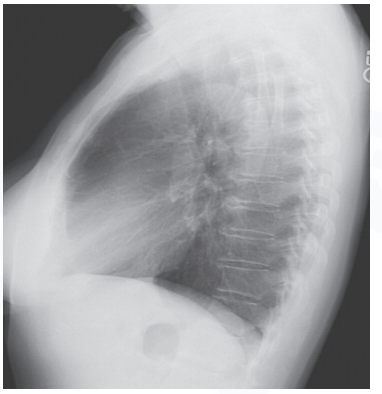
Most dependent part of Pleural cavity in Erect position is posterior recess

CXR- PA view



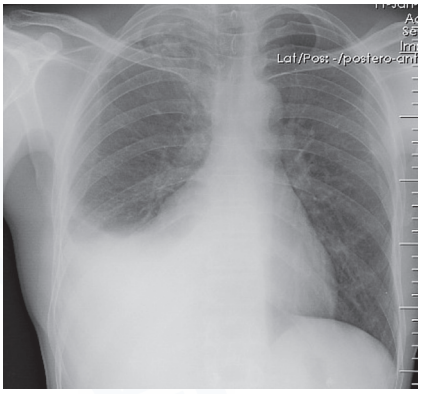
Part of Pleural Recess seen On CXR- PA View is NOT posterior Recess

CXR- Lateral View



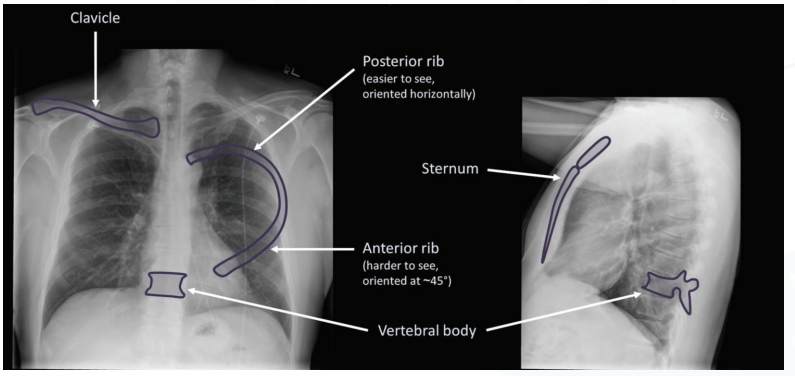
Part of Pleural Recess seen On CXR- Lateral View is

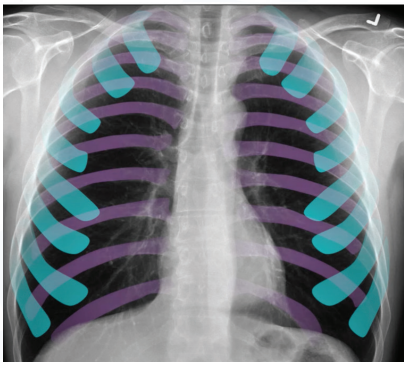
CXR- PA view



Amount of fluid needed for Blunting Lateral CP angle on CXR PA viewQ

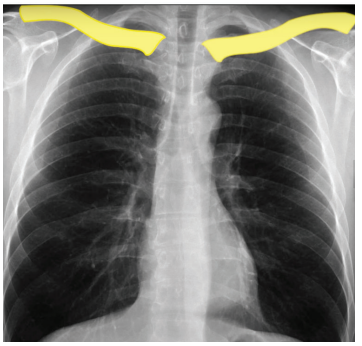
Bones on CXR





Q. Anterior End of Rib ?

Q. Posterior End of Rib ?

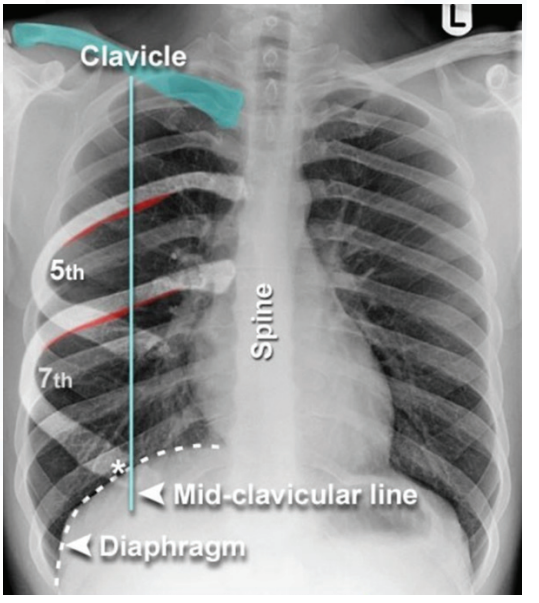


Rib counting : Concept

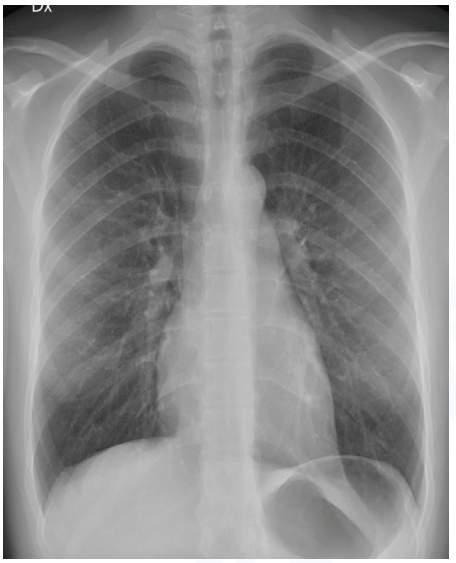
Key Concepts in Rib Counting :

1. Posterior end of rib – Directly attached to
   Lateral Vertebral Margin
2. Anterior end of Rib – Indirectly attached to
   Sternum – Through
   Costal Cartilage
3. Costal Cartilage is NOT well seen on X-ray
4. One part of Rib will reach Close to Midline
   Other part of Rib will remain Laterally
5. First Rib is C-shaped structure overlapping Clavicle

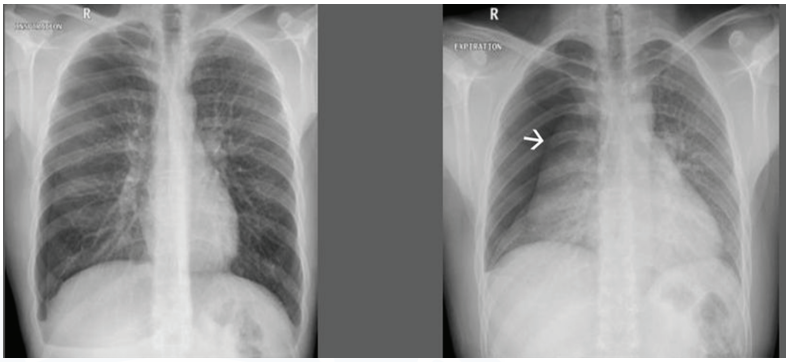
Count the Ribs



CXR is PA, erect and inspiratory

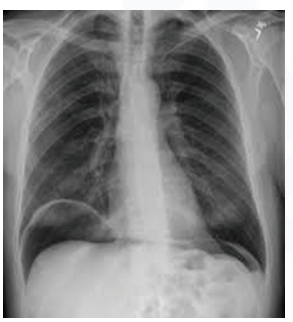


Expiratory CXR is preferred in-



Workbook Questions

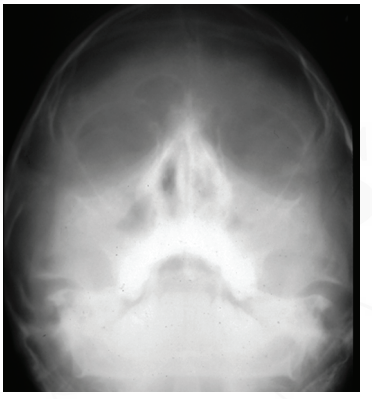
Q.1 Elderly pt. taking analgesics for knee O.A. Pt. experience sudden onset abdominal pain. CXR is done in ER and shown below ?



Q.2 Identify the abnormality shown below?

Q.3 Identify the abnormality shown in the below image?

Q4. Young adult with repeated chest infections & Sinusitis? CXR done is shown below. What is the likely diagnosis?



Q 5. Best CXR for Pneumothorax is?

1. CXR – Supine Inspiratory
2. CXR - Supine Expiratory
3. CXR- Erect Inspiratory
4. CXR - Lateral

Mammography: Imaging of breast

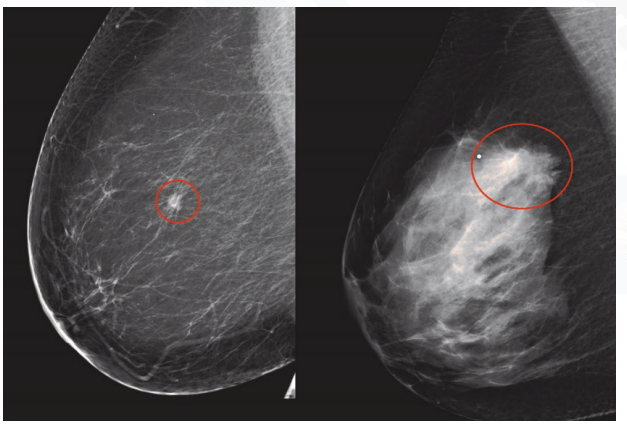
Two views-
Mediolateral Oblique
Cranio-caudal

Which view shows ?Q
-More tissues
-More Important
-Includes Axillary Tail

- Mammography – with X-rays
- Sono mammography – with USG
- MR Mammography- with MRI

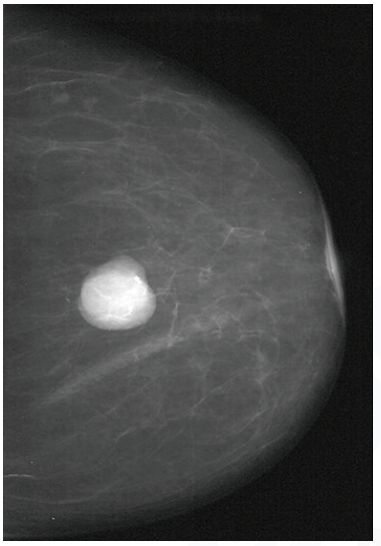
When to do a mammography?
Age of the pt

Elderly Female Young Female



- With increasing age- Gland converts into fat
- Gland is white n Breast Mass is also white.
- Ability to detect Breast Mass / Cancer is more in elderly or young ?

Benign vs. Malignant lesion Q



Well defined Rounded mass

OR

Ill-defined Irregular **Spiculated Mass**

Architectural Distortion Absent

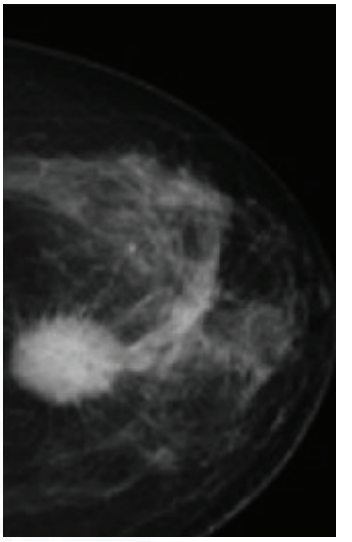
OR

Architectural Distortion **Present**

Calcification Macro-

OR

Calcification **Pleomorphic Micro-**



Calcifications...

Macrocalcification (> 2mm)

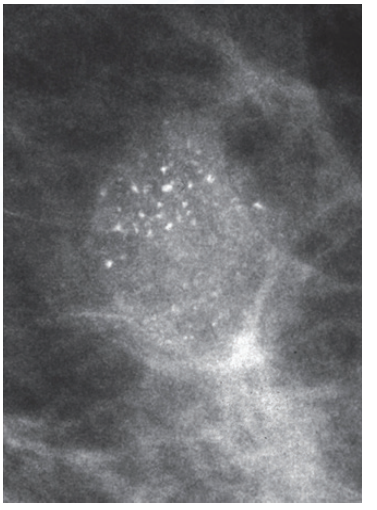
Microcalcification(< 0.5 mm)

Popcorn Calcification

Brain

Breast

CXR



Imaging Reporting and Data System .....

Tells about Risk of disease – NOT severity

- BI RADS - Breast Imaging Reporting and Data System (Risk of Cancer) Q
- TI RADS - Thyroid
- LI RADS - Liver
- PI RADS -
- CO RADS – COVID Pneumonia

Grade	Interpretation	Managemant
0	Incomplete evaluation	Complete it
1	Normal	None
2	Benign	None
3	Probably benign (<2%)	Short follow-up
4	Suspicious/ indeterminate	Biopsy
5	Highly suspicious (>95%)	Biopsy

Q. If a palpable mammographic lesion is noted and there is low index of suspicion (>2%), CORRECT further advise should be?

1. Mammographic follow-up annually
2. Mammographic follow-up in 3–6 months
3. Stereotactic core biopsy
4. Surgical biopsy

Screening Mammography	Diagnostic Mammography

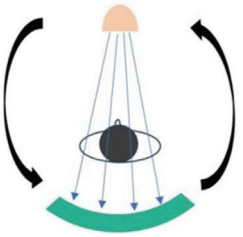
Breast Investigation in Special Situations Q

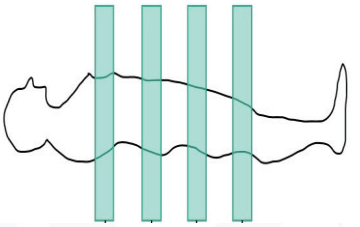
1. Breast Lump in Pregnant female-
2. Breast Lump in Lactating Female-
3. Ix.O.C. in female with Breast Implant-
4. Ix.O.C. to differentiate between post surgical scar and recurrent breast cancer-

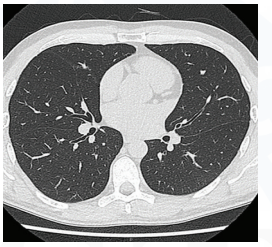
CT: Principle

CT also uses X-ray Tube which is MOVING & two important difference with radiography is

1. Exact localization of the structure within the body







2. Different soft tissue can be differentiated- as they have different CT number (H.U value)

Quantifying the Radio densities seen :

HU Scale/Value

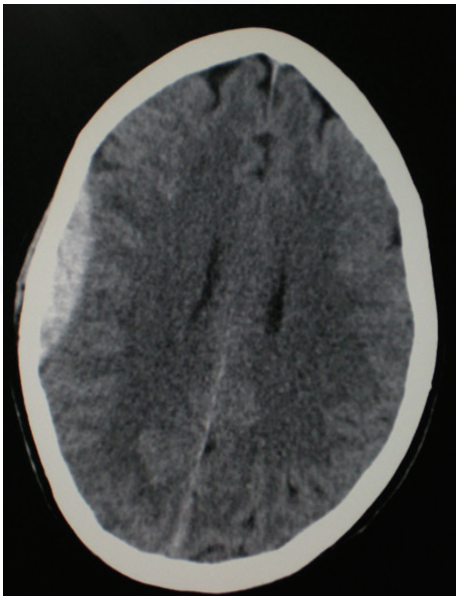
+ 1000 HU- Cortical Bone

+ 100 HU - Acute Hematoma

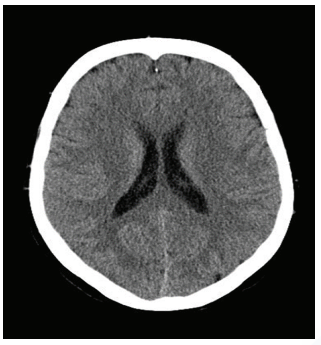
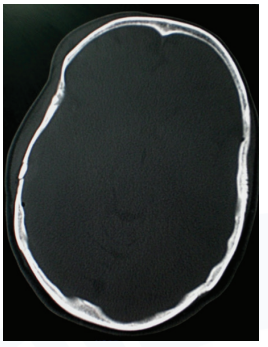
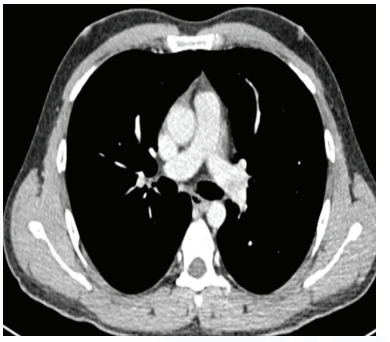
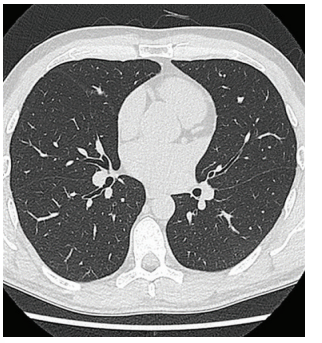
0 HU - Water

- 100 HU – Fat

- 1000 H.U. - Air



CT window

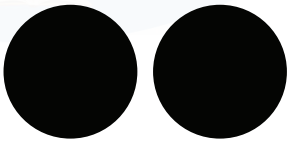
Q. Highest HU value is of which pathological condition ?

1. Pneumocephalus
2. Simple Renal cyst
3. Acute Extradural Hematoma
4. Lipoma

Applications: CT Scan

- Acute haemorrhage in Brain
- Cortical bony pathologies – Simple vs. Com plex #
- Calculus/Calcification
- Free air anywhere

- All trauma except Head Trauma and Unstable Pt.
- Lymph node Characterization- Necrosis +/- Conglomerated Necrotic Lymph node-

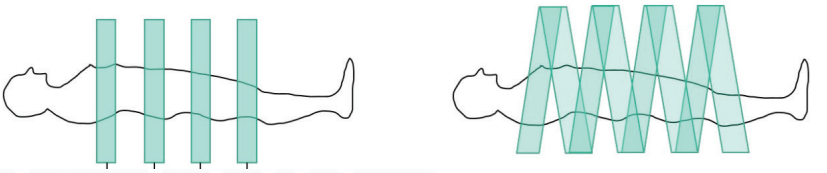


- Staging of tumour -
- Infective/inflammatory conditions
- Superficial Abscess
- Deeper Abscess

HRCT (High Resolution CT) : Non Contrast CT

Indication

- Pulmonary



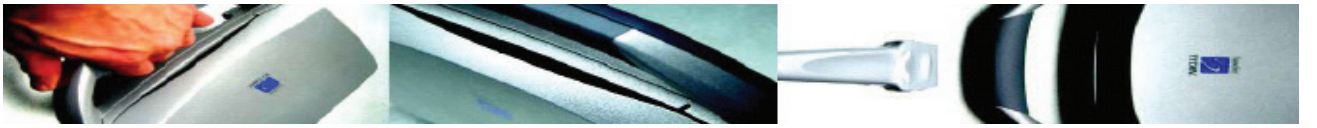
- Extra pulmonary

Q. Investigation done prior to CECT study?

1. KFT
2. LFT
3. Blood Sugar
4. Thyroid Function Test

Q. Which of the following investigation can safely be done in a patient with Renal failure?

1. CE CT
2. CE MRI
3. CE USG
4. Angiography



Ultrasonography & Doppler

Principle-

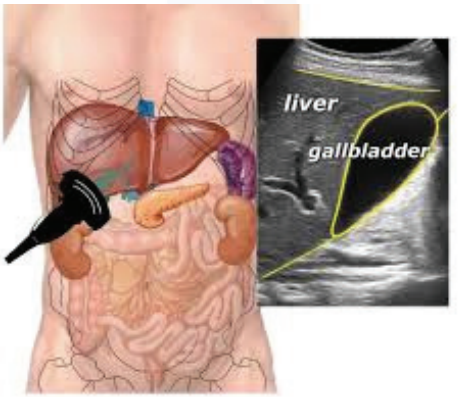
Probe crystal –

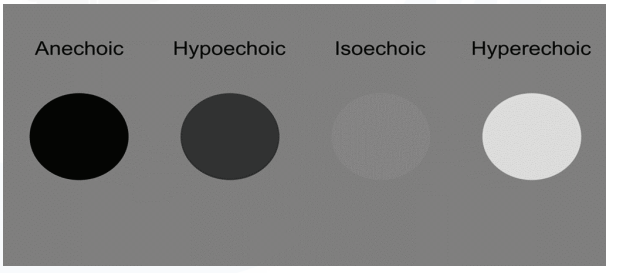
Merits and Demerits: USG

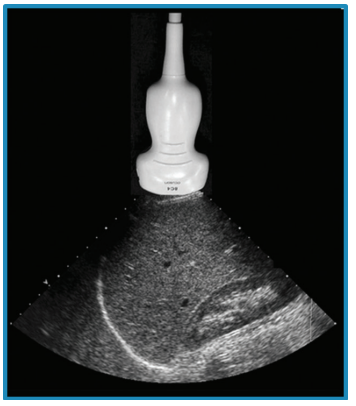
- Low Cost / Easily available
- Non-Ionizing Radiation –Safe in pregnancy
- Non-invasive
- Portable and Fast
- Real Time and Dynamic
- POCUS – Point of Care USG
- Highly Operator Dependent
- Poor Penetration in Air and Bone

Q. Patient has met with RTA. With low BP and tachycardia the pt. is brought to the ER. Next step in the Mx.

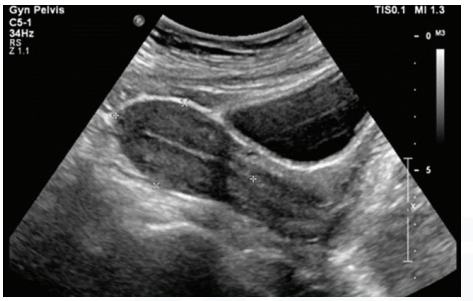
USG: Terminologies



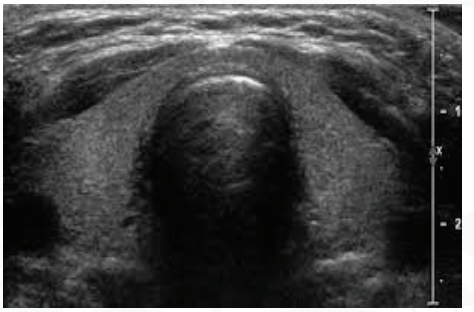




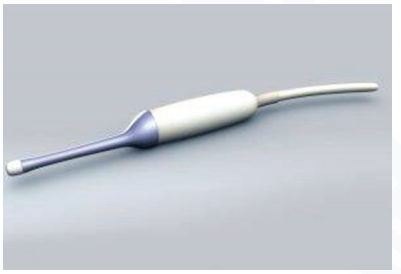
Choosing the right probe

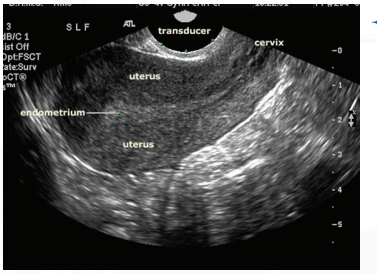


- **Curvilinear-**Curved Probe surface
  - ⇨ Low Frequency
  - ⇨ Depth
  - ⇨ Resolution

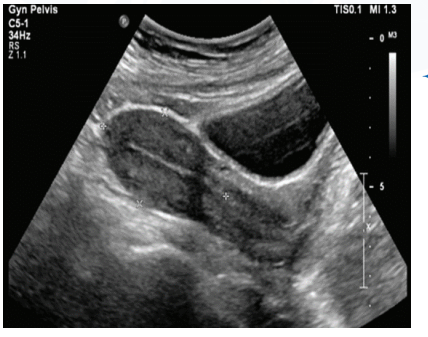


- **Linear-**Flat Probe surface
  - ⇨ High Frequency
  - ⇨ Depth
  - ⇨ Resolution





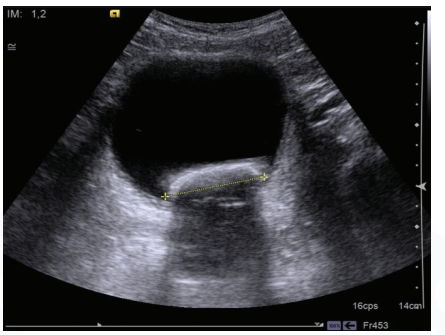
- **Intra-Cavitary Probe**
  - ⇨ TVS
  - ⇨ TRUS

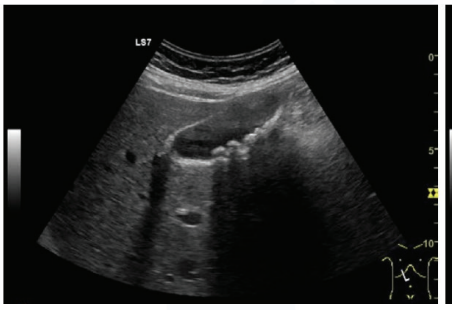


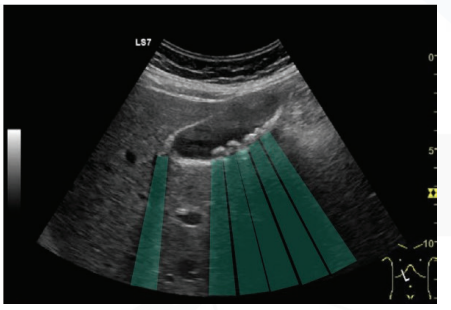
- **Curvilinear Probe**

Posterior Acoustic Shadow: If a calculus comes in the path of USG beam, there is reflection of beam and area behind it appears dark.

This is diagnostic for

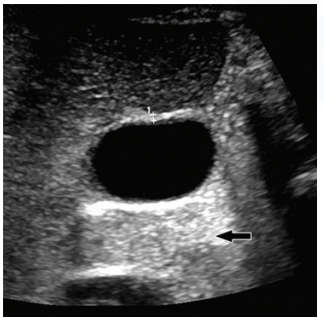


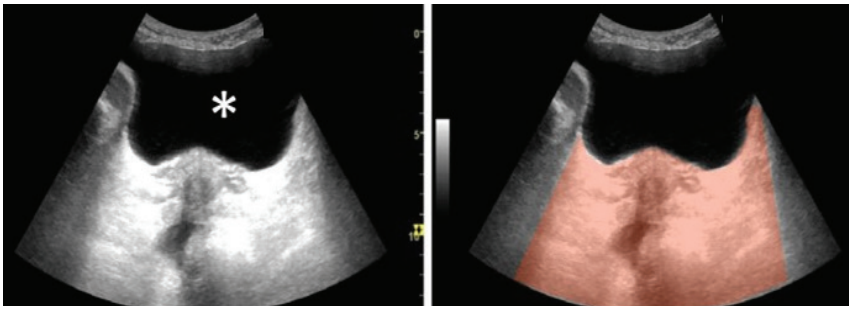




Posterior Acoustic Enhancement: If a fluid filled structure comes in the path of USG beam there is complete passage of the beam and the area behind it appears brighter.

This is diagnostic for

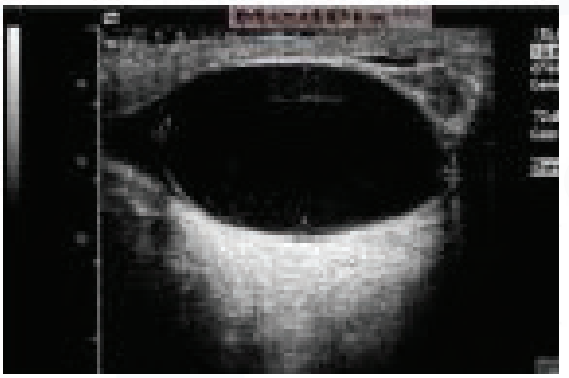




Q. All of the following appears hyper echoeic on USG except ?

1. Air
2. Fat
3. Bone
4. Bile

Q: The USG image shows a phenomenon seen in?



Q Acoustic shadow is produced by ?

Applications of USG

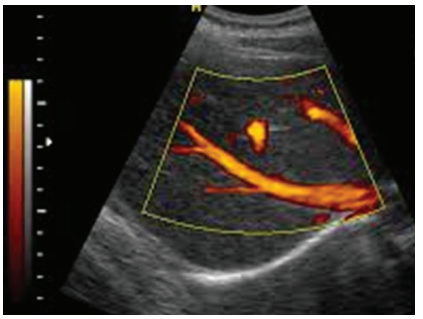
1. Intra- cranial : Trans-fontanelle USG up to age of
2. Ocular USG
3. Neck : Thyroid
   Lymph nodes
   Carotid Vessel
4. Heart : Echocardiography
5. Breast:
6. Thorax: Pleural effusion / Pericardial effusion

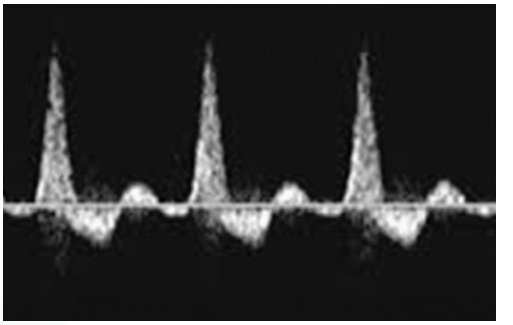
4. **Abdomen:**
   GB pathology
   CBD calculus
   Renal colic
5. **Pelvic** pathology
6. **Pregnancy**
7. **Scrotal pathologies:** Inv. Of Choice for all except

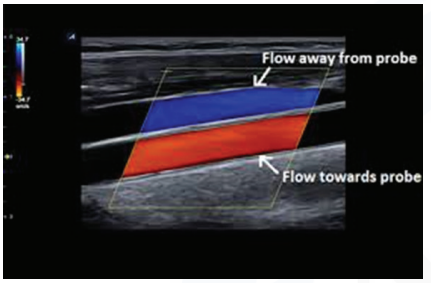
Q. All of the following are indications of USG in emergency except?

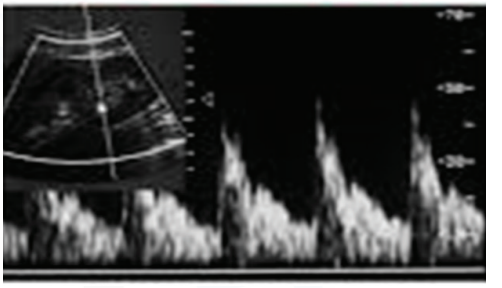
1. Diagnosis of Cholecystitis
2. Diagnosis of free fluid in Trauma patient
3. Diagnosis of Acute Pancreatitis
4. In a patient with suspected pulmonary thromboembolism to r/o DVT

Doppler USG





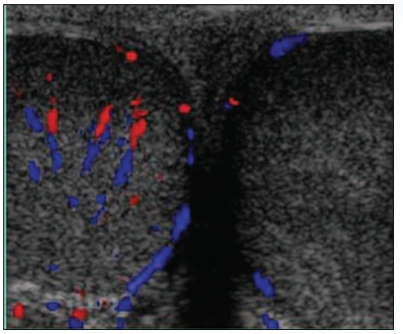




USG Doppler Investigation of choice in:

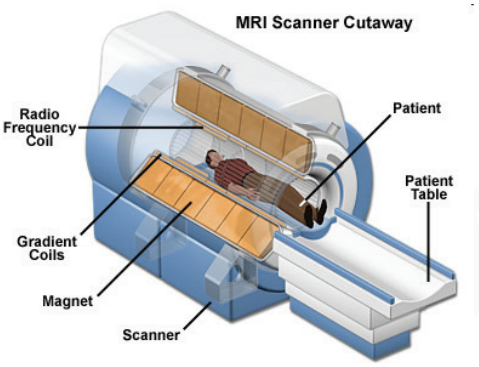
- DVT
- Varicose veins
- Scrotal Varicocele
- Testicular torsion

Q. Acute onset scrotal pain . USG image is shown below . What is the diagnosis?



MRI (Magnetic Resonance Imaging)

CT is a Ring vs MRI is a cylinder



- MRI is a non-ionizing, non-invasive imaging modality
- Long Noisy examination
- Principle- Gyromagnetism Q
- Radiation Used-
- Faraday's Cage- Copper shielding of MRI room
- Contrast Used-
- Strength of MRI –

MR Safety – Absolute C/I Q

- Free lying intra-ocular ferromagnetic foreign Body
- Retained Bullet inside Body
- Pacemaker but not coronary stent
- Aneurysmal Clips
- Cochlear implant but not Ortho or Breast or Dental Implant
- Metallic Cardiac valves
- Any magnetic device in Body but not Cholecystectomy clips/IUCD

- TWO Relative Contraindications: Clautrophobia / 1st Trimester Pregnancy

Applications: MRI

- Neural tissue

  Central Nervous System-

  Peripheral Nervous System-

- Cancellous bone - Osteomyelitis / Avascular Necrosis / Staging of Bone tumour

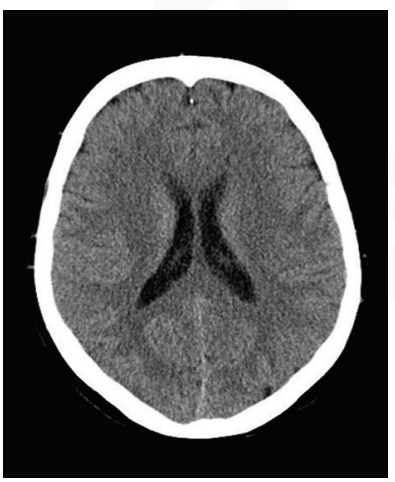
- Cartilage - Chondromalacia / Developmental Dysplasia of Hip

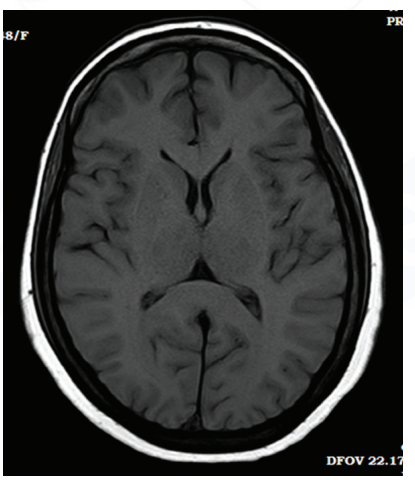
- Ligaments and tendons

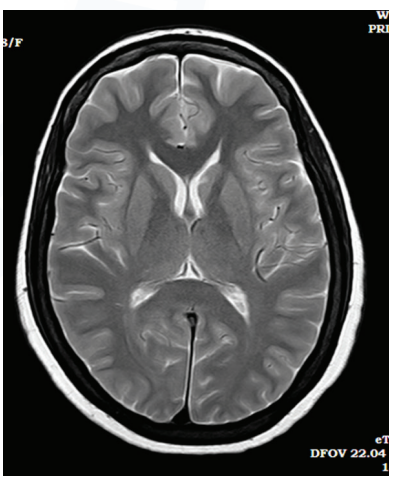
Terminologies in MRI

Identify the image – CT / MRI

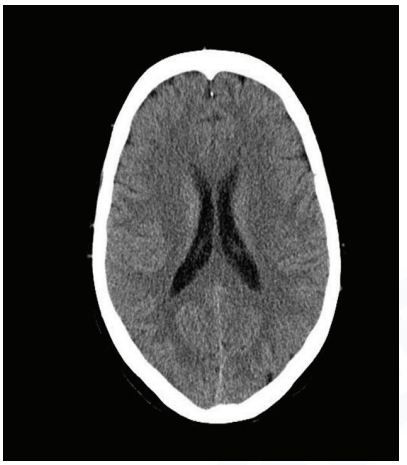
Cortical Bone – White or Dark

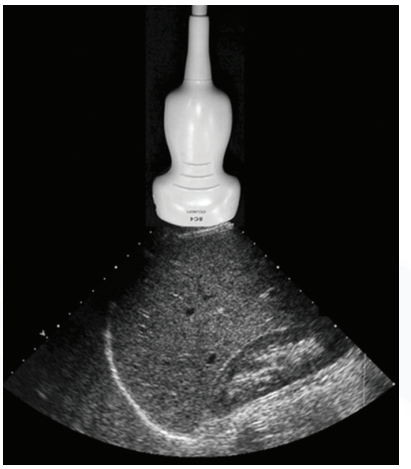


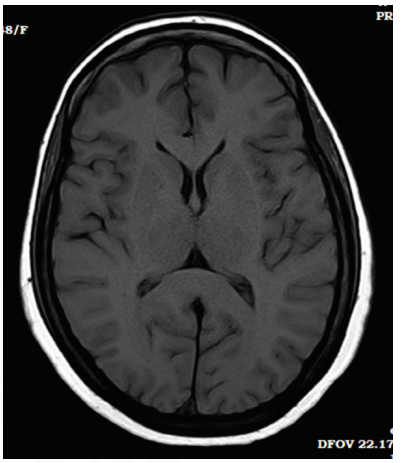




Terminologies in Radiology Investigations







Referral for Radiological Investigations

- Test utilizing Ionizing vs. Non-ionizing radiation – Non-ionizing investigation is preferred
- Invasiveness of the test – Non-invasive over invasive test
- Clinical scenario in which the test has to be done

Investigations in Radiology

Radiological Scans (Uses X-rays)	Nuclear Scans (Uses Gamma rays)	Investigations using Non-Ionizing Radiations
Radiography/CT	Scintigraphy/SPECTPET	
Mammography	DTPA/DMSA/MAG-3	
IVU/MCU/RGU	Pertechnate	
Angiography	MIBG/Sestamibi	
DEXA	HIDA	
	Bone Scan	
	Octreotide Scan	

Q. Which of the following investigation does not use Ionizing Radiations?

1. CT Scan
2. MRI
3. Bone Scan
4. Mammography

Q. X-rays are not used in which investigation?

1. DSA Scan
2. DEXA scan
3. CAT Scan
4. Bone Scan

2. Invasiveness of the test

- Invasive Test- Breaking the natural protective barrier

  Eg: Laparotomy / Laparoscopy

- Minimally invasive : Biopsy / Intra-arterial injection (Catheter Angiography)

- Micro-invasive : CECT / CT Angiography

- Non-Invasive test: USG

3. Types of test depends on Clinical Scenario

- 1st Investigation
- Screening Test – To r/o the ds.
- Investigation of Choice – To confirm the ds.
- Gold Standard test – Final word
- Best Investigation

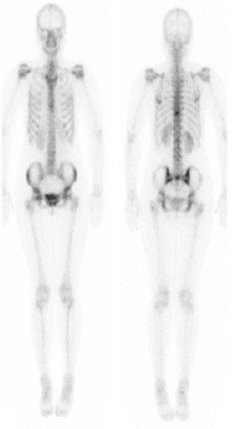
Screening test vs. Ix.O.C ?

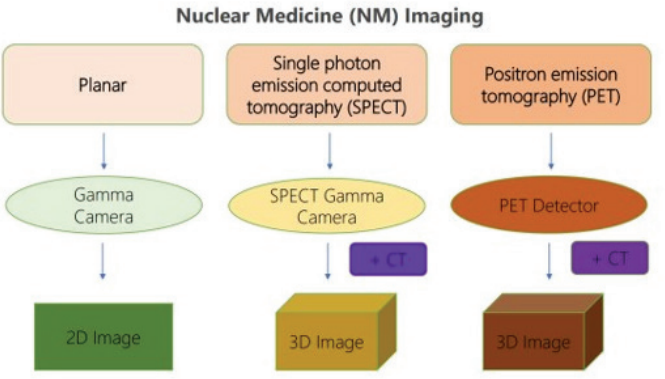
- Risk Factors +/-
- Symptoms +/-

Next step/investigation...

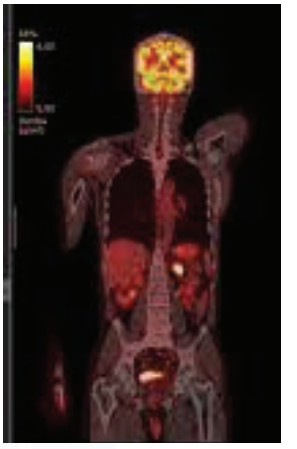
NUCLEAR SCANS

PLANAR SCINTIGRAPH

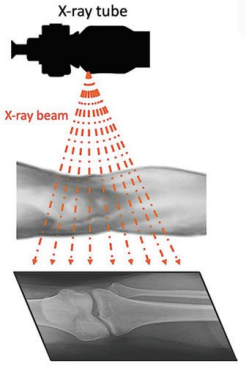


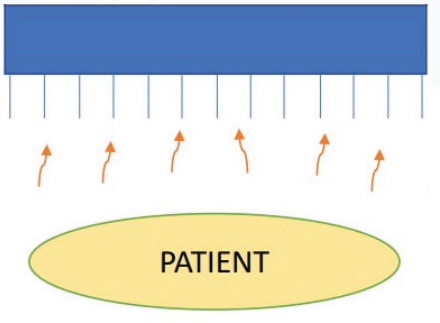


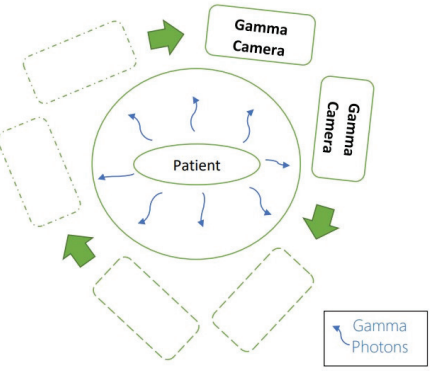
PET SCAN



Radiography - Planar Scintigraphy - SPECT

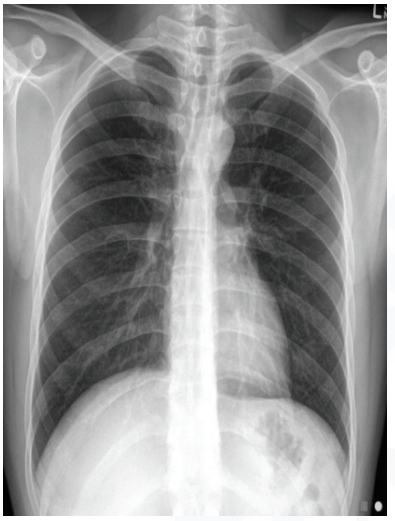




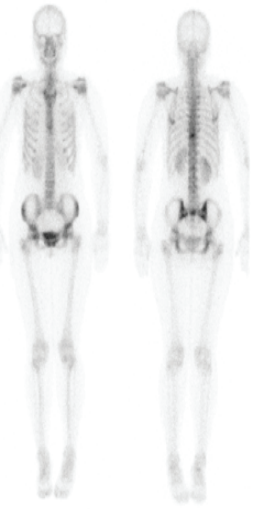


Difference b/w Radiological and Nuclear scan

Anatomical : CXR



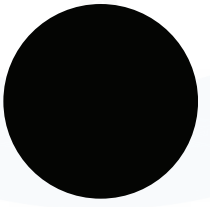
Functional Scan : Nuclear Scan

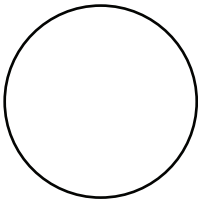


Interpretation of Nuclear Scan: Hot spot vs. Cold Spot

Radiological Scans

Nuclear Scans





Example of Nuclear Scans: Tc 99 – DMSA Scan

Tc99 pertechnate scan Q

- Gastric Mucosa –for ectopic gastric mucosa in Meckel's Diverticulum
- Salivary Glands – In Warthin's tumour
- Thyroid Gland - In India Pertechnate Scan is used for Thyroid Imaging

Radioactive Iodine Uptake Scan (RAIU) – Iodine

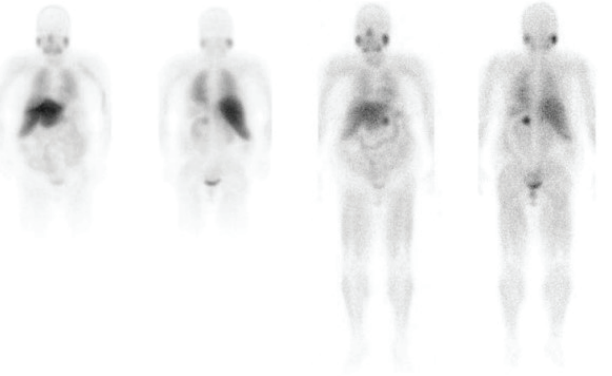
Radioactive Iodine Ablation (RAIA) – Iodine

Commonly used Nuclear scans[Q]

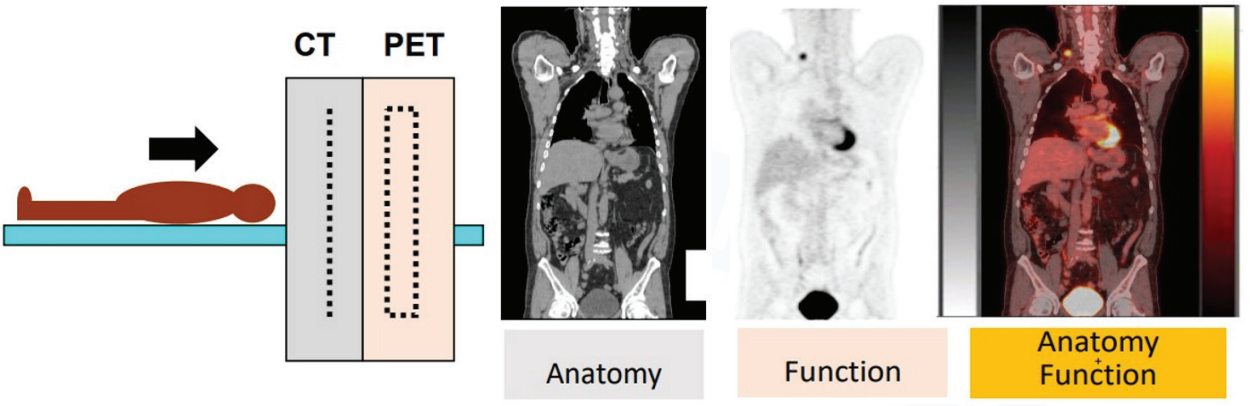
Radionuclide	Half-life	Pharmaceutical Agent	Clinical Indications
Technetium-99	6 hrs	Pertechnate[Q]	Thyroid / Meckels scan , Warthins Tumour
		MDP	Bone Scan for osteoblastic mets.
		Sestamibi[Q]	Cardiac , Parathyroid
		MAA (Human Albumin)	Lung Perfusion
		DMSA[Q]	Renal Scar
Iodine-123	13 hrs.		**Thyroid Scan**
		MIBG[Q]	Pheochromocytoma/ Paraganglioma
Indium-111	2.8 days	Octreotide Scan [Q]	Neuroendocrine Tumours
Krypton-81	13 seconds	Lung Ventilation scan	PTE

Q. Figure demonstrates a left Pheochromocytoma. What nuclear medicine study do these images represent?

1. Gallium-PSMA
2. FDG PET Scan
3. I123-MIBG
4. Tc-Pertechnate



Positron Emission Tomography (PET) Principle



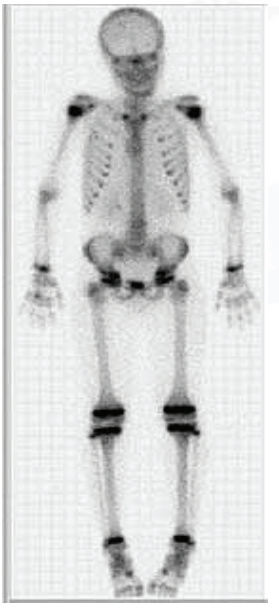
Commonly used Positron emitters in PET Scan[Q]

Flurodeoxyglucose (FDG) [Q]	110 Minutes	Glucose Metabolism	Oncology, Infection
Gallium-68 Dotatate Scan[Q]	68 minutes	Somatostatin Receptor Expression	Neuroendocrine tumours
Gallium-68 PSMA ScanQ	68 minutes	PSMA expression	Prostate Cancer

FDG PET Scan

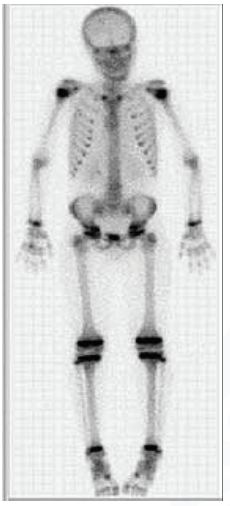
- Blood test prior to PET Scan
- Uses PET Gantry and NOT Gamma Camera
- Normal tissue showing activity on PET
- Marker of -
- Disease mimicking cancer on PET
- Cancer which do not show PET activity -

Scan Type: Site of Radionuclide Uptake

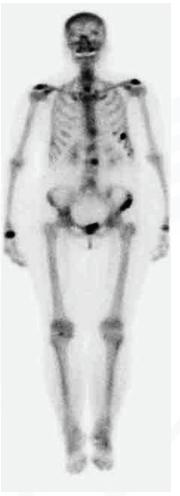


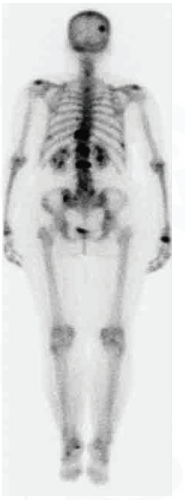
Normal vs. Abnormal

Symmetric



Asymmetric Activity on Bone Scan:





Thyroid Scan
Diffuse Inc. Activity
Focal Inc Activity
Cold Nodule

Bone scan is not useful in -

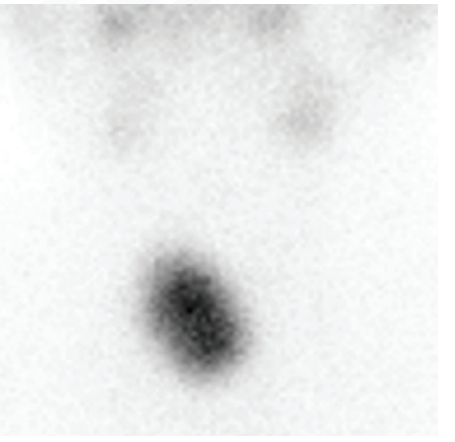
Q. 36 years old female patient presents with clinical and biochemical symptoms of hyperthyroidism. As part of the investigation, she had a [99mTc]Tc-pertechnetate thyroid scan (shown below). What diagnosis is demonstrated in the study?

A. Graves Disease

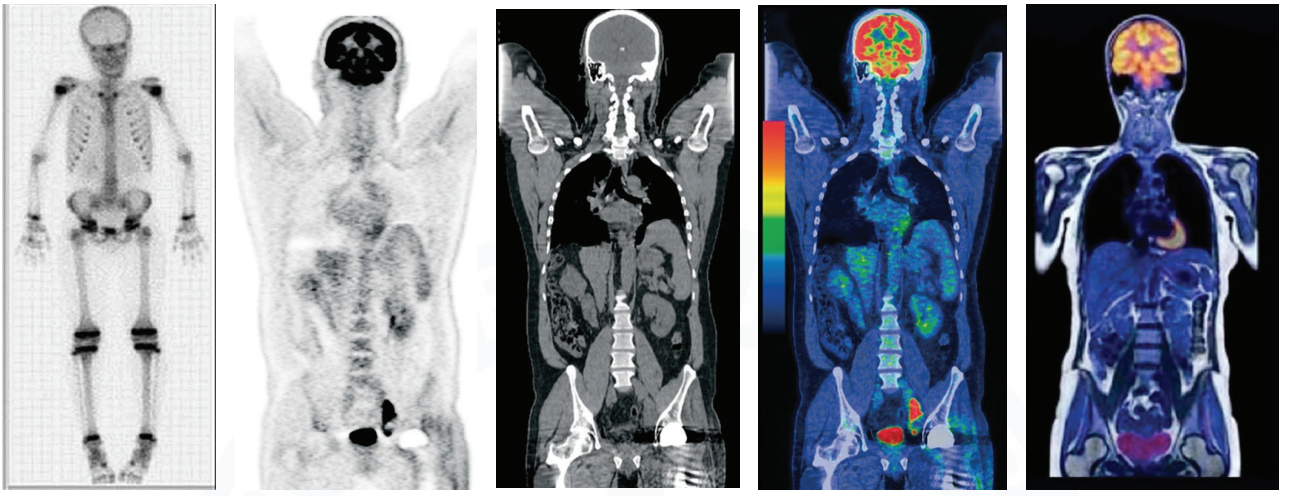
B. Toxic MNG

C. Toxic Nodule

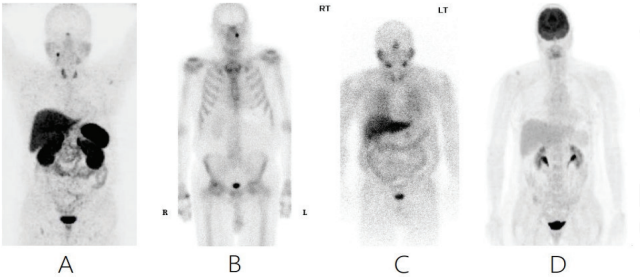
D. Subacute Thyroiditis



Nuclear Scans

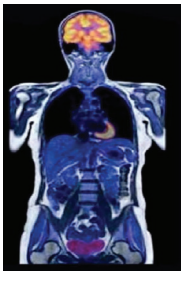


Q. Which of the following image represents FDG PET Scan?



A B C D

Q: Identify the imaging modality?



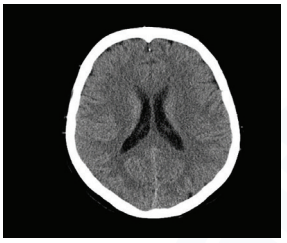
1. CT Scan
2. MR Imaging
3. Fusion Imaging
4. Angiography Imaging

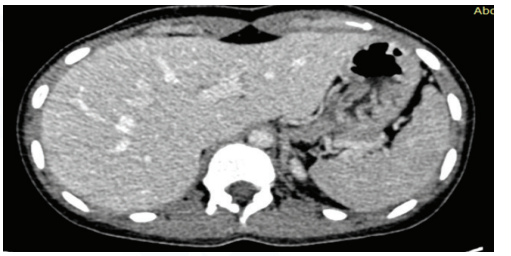
Q. Which radionuclide is PSMA labelled for the detection of metastatic disease in prostate cancer ?

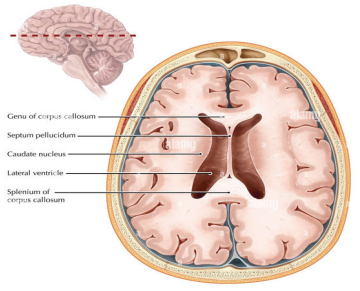
1. Gallium-68
2. Rubidium – 82
3. Tc-99
4. Iodine -123

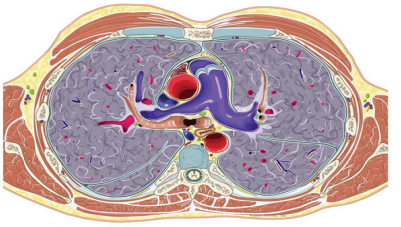
✦ Cross-Sectional Radio- anatomy

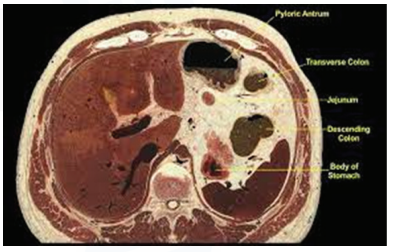
1. Identify the body part



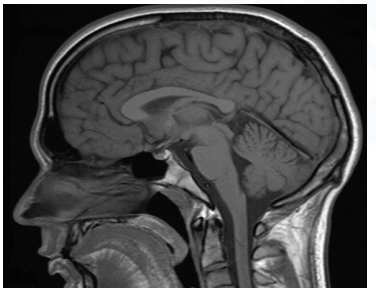


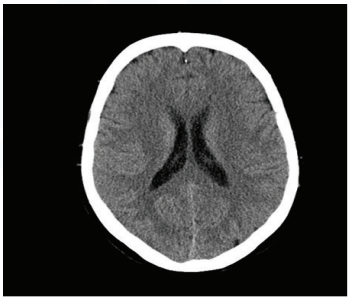


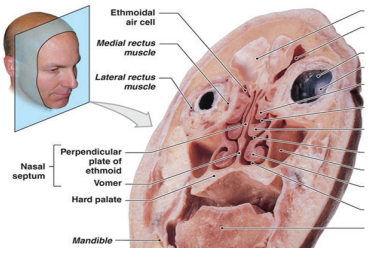


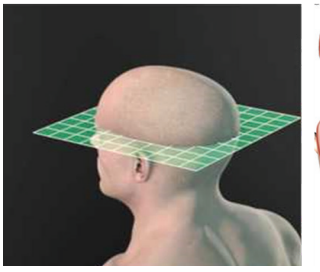


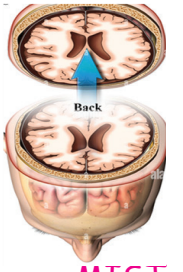
2. Identify the plane of section



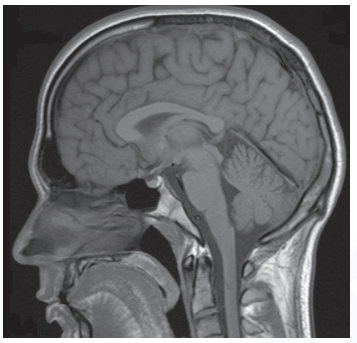


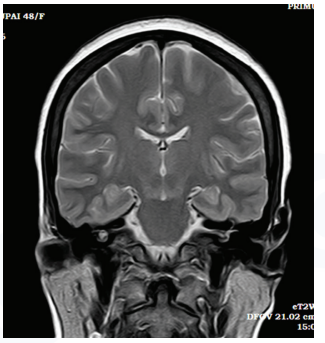


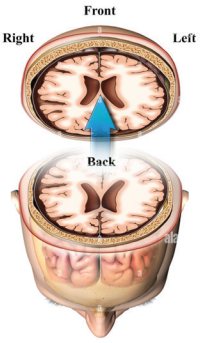


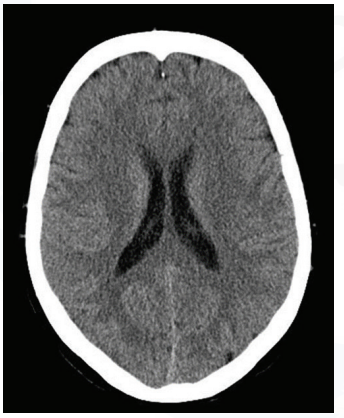


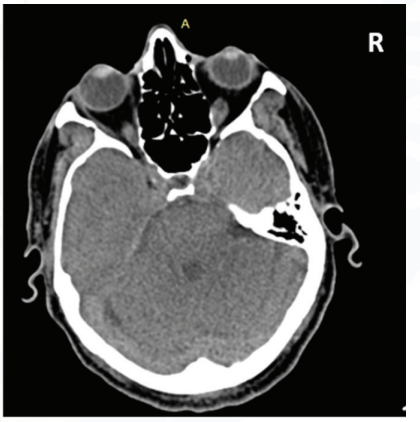
3. Right-Left Orientation





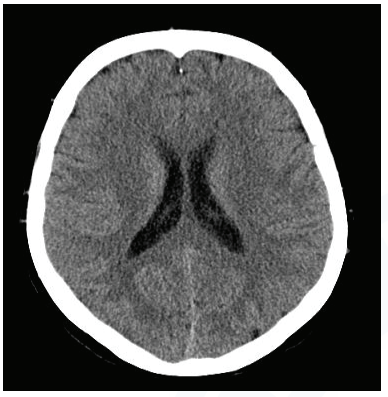
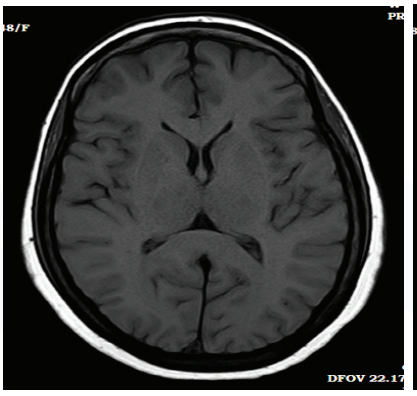
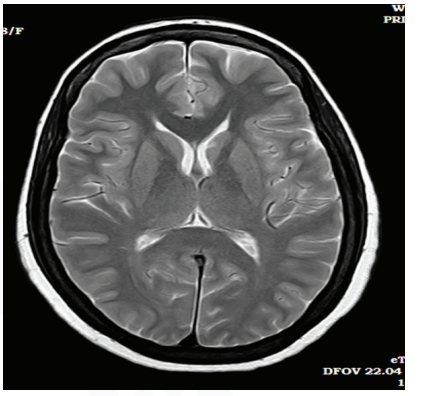




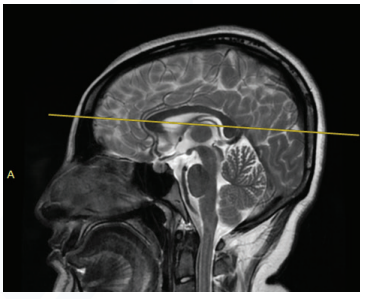


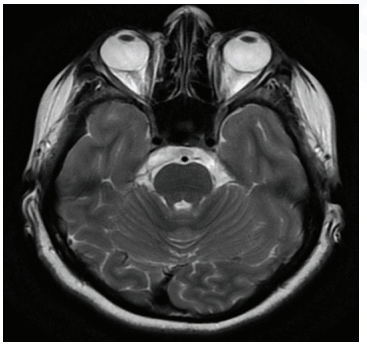
4. Modality

Q. CSF appearance in the image..

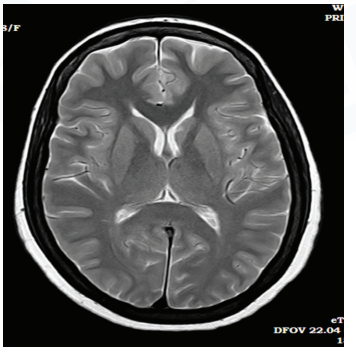




Anatomical landmark in brain- Lateral Ventricle

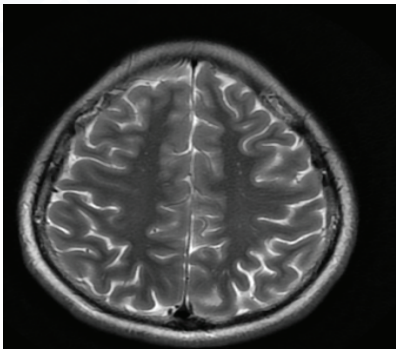




Infra Ventricular

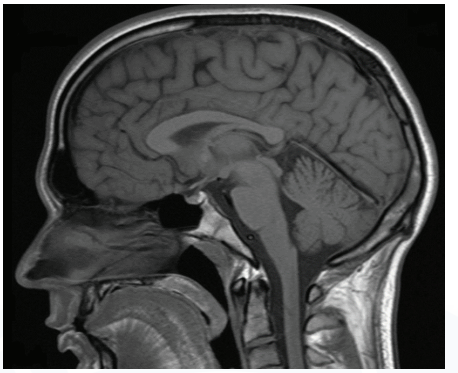


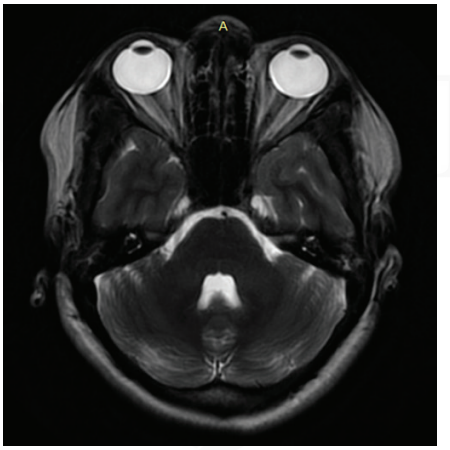
Ventricular Level

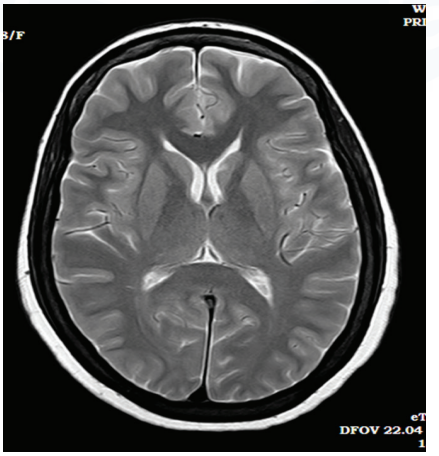


Supra-Ventricular Level

Q. Structure identification Q

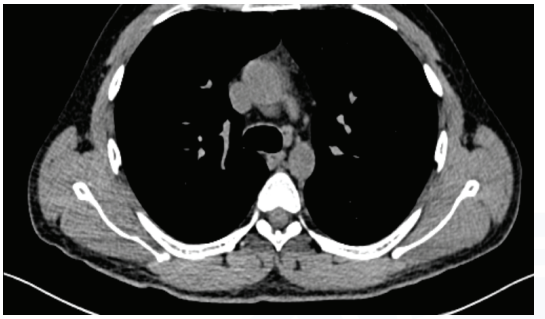






NCCT vs CECT

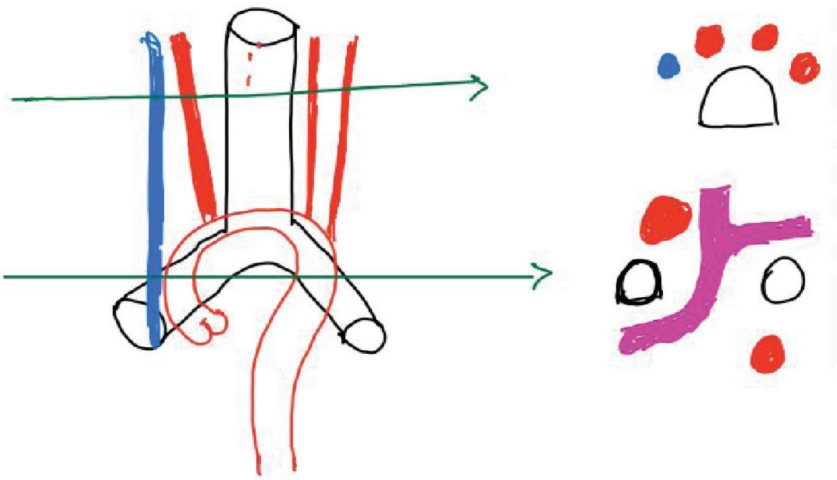
If Vessel is same as Muscle



If Vessel is whiter than Muscle

Never compare vessel with

Thorax Landmark: Trachea

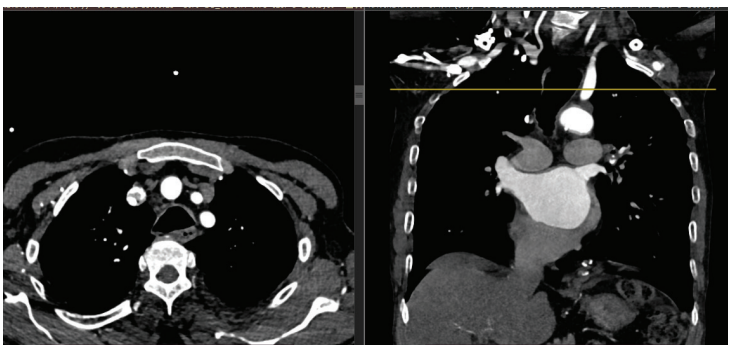


Landmark for evaluation in Thorax: Trachea

If Trachea is seen which appears as single black tube -

Level is **Above carina and arch**

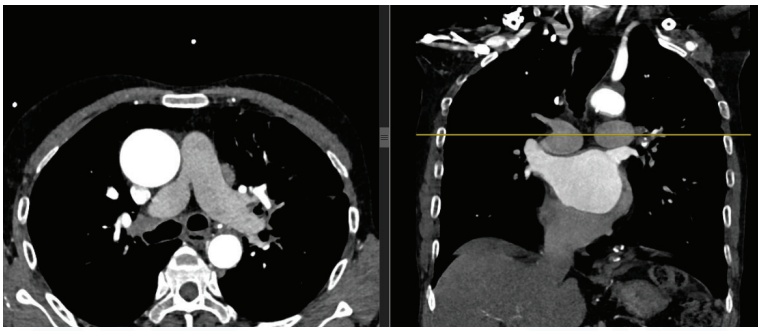
The vessels seen are



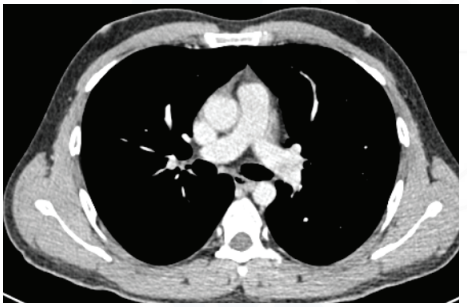
Trachea bifurcates into Rt and Left Main Bronchus- two air tubes

Then **Level is Below the carina and arch**

Vessel is below the arch i.e.

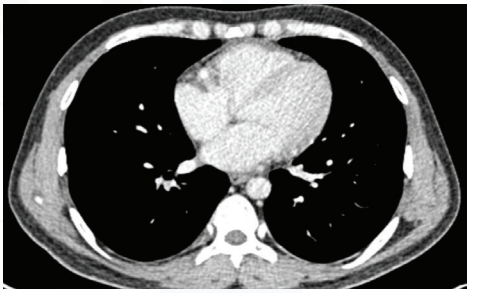


Pulmonary artery Q

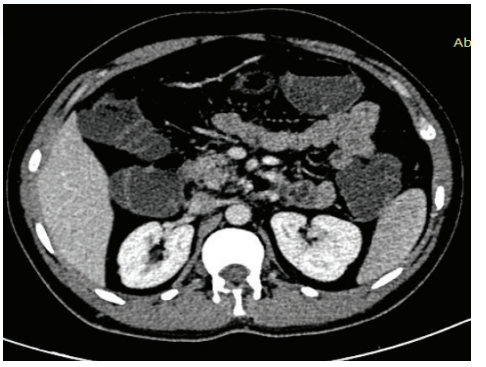


Cardiac Chambers Q

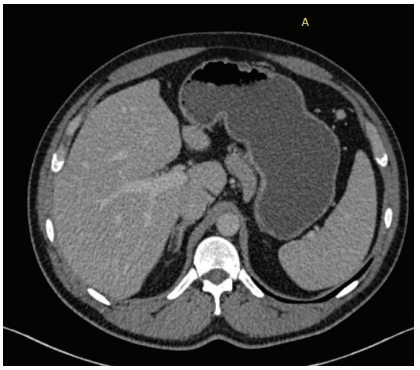
LA- Posterior most chamber of Heart

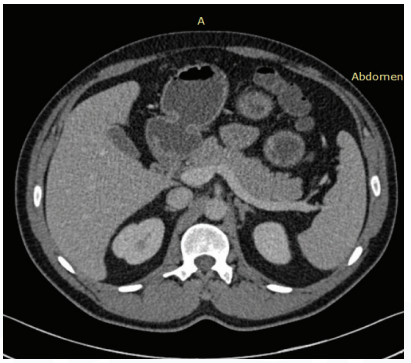


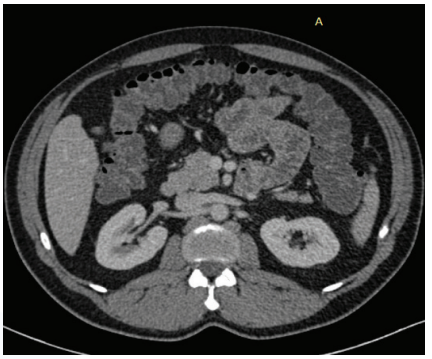
Landmark in Abdomen is Kidneys



Structure Identification Q



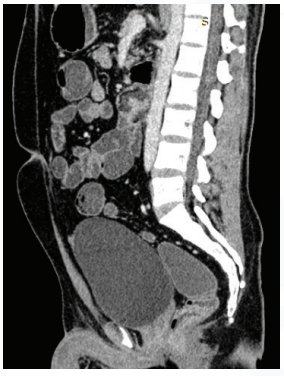


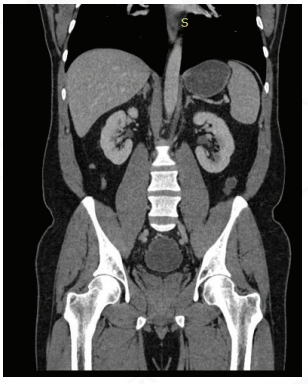


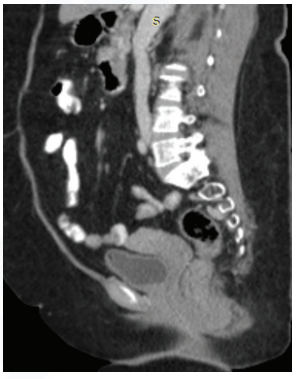
Male vs Female Pelvis:

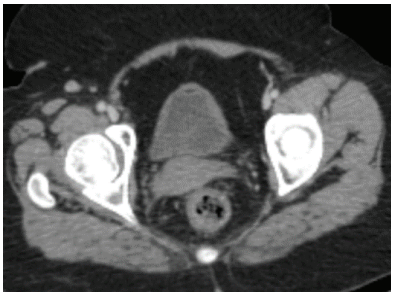
The prostate lies below the

Uterus lies behind the

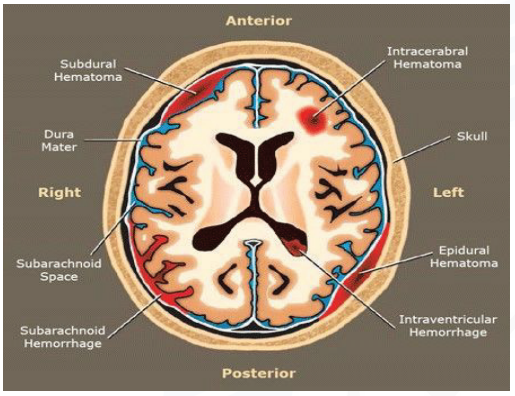


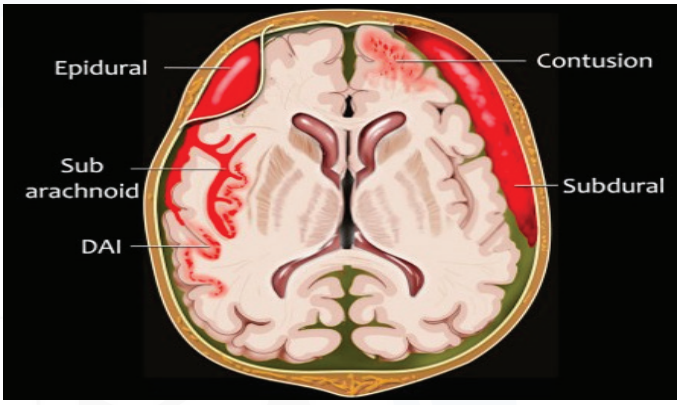






Head Trauma Q

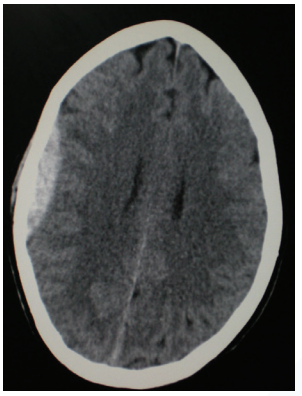
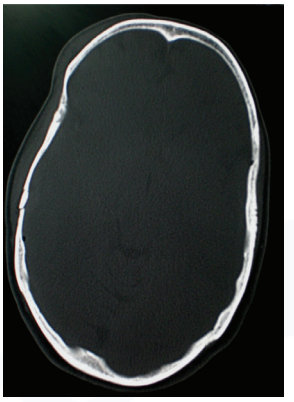
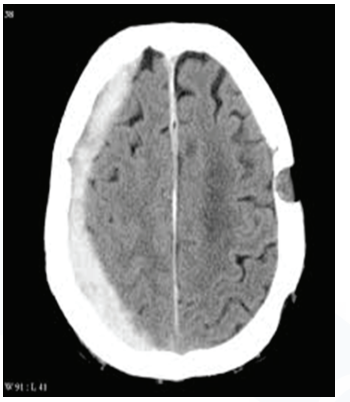
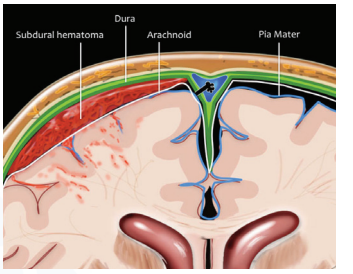




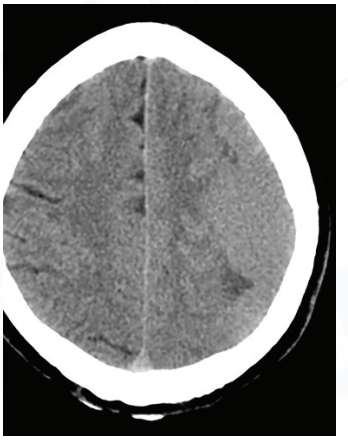
Patient with Head Trauma- 1st Investigation: NCCT

Extradural / Epidural Hematoma Q	Subdural Hematoma (SDH) Q
✦ Biconvex/ Lentiform /Idli ✦ Significant injury- Acute Presentation ✦ Lucid Interval ✦ Arterial Bleed- M/c vessel ✦ Does not cross sutures ✦ A/W Skull fracture	✦ Concavo-convex/ Crescent / Banana ✦ Significant injury (Acute) to Insignificant injury ✦ No Lucid Interval ✦ Venous – Bridging cortical veins ✦ Can cross sutures ✦ Not a/w skull fractures

Imaging

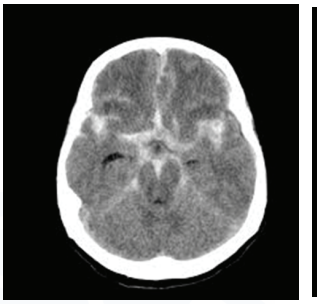





Different Stages of Subdural Hemorrhage Q



Subarachnoid Haemorrhage Q

- Most common cause:
- Most common cause of spontaneous SAH :



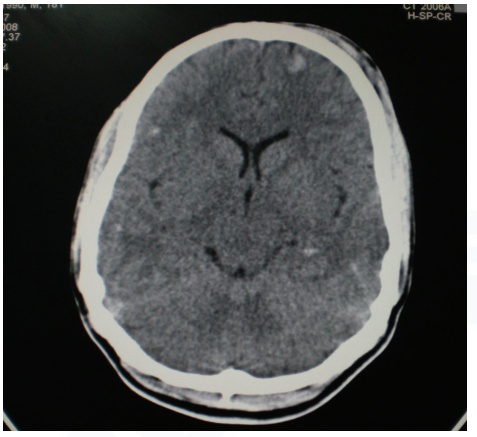
- Shape
- Blood in CSF spaces-
  Ventricles / Fissures / Cisterns & Sulci
- Presentation :

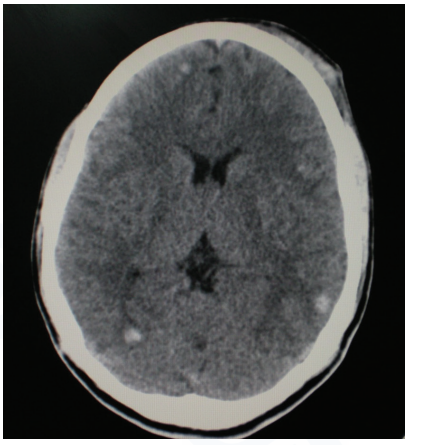
Q. Young female with Headache & loss of consciousness. NCCT done in ER is shown below.

- What is the diagnosis
- What is the cause
- What is the vessel involved
- What is the next step
- Gold Std. for diagnosis
- Two treatment options-
- Preferred treatment-

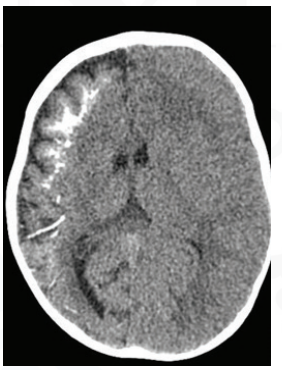
CT findings in DAI

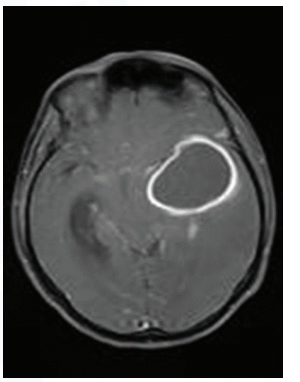
Small hemorrhage in Gray-white junction / Corpus Callosum / Brain Stem





Q. What is the diagnosis in following four images:



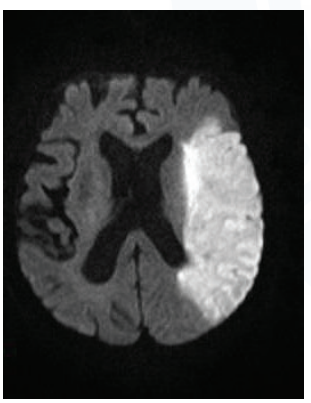


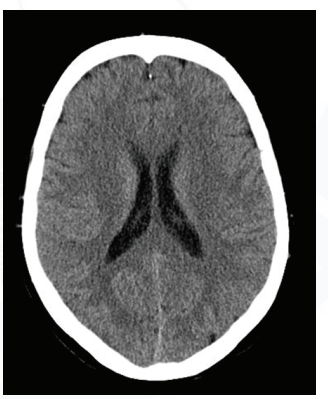
1. SAH
2. Tumour
3. Calcification
4. Tumefactive Demyelination
5. Bleed
6. Calcification

Q. Young male brought unconscious to hospital with external injuries. CT brain is normal. What is the diagnosis?

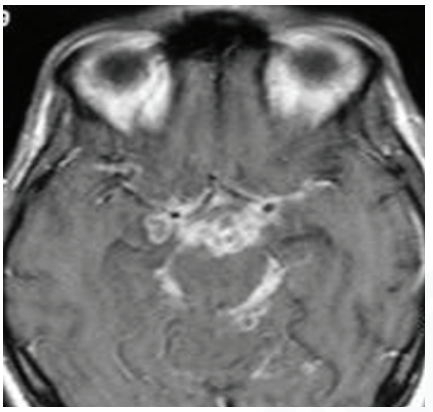
1. Cerebral contusion
2. Cerebral laceration
3. Multiple infarcts
4. Diffuse Axonal Injuries

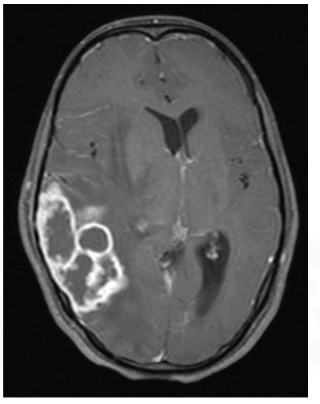
Pt. with suspected Stroke Q



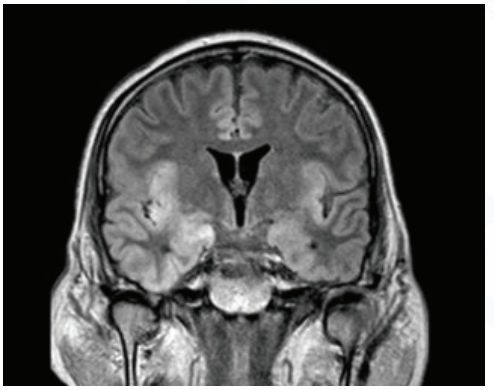


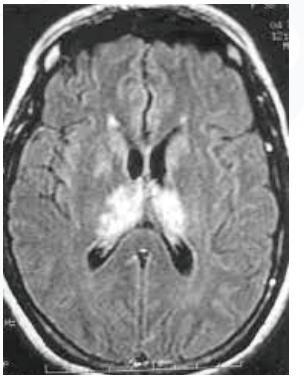
Patient with fever + Neck rigidity Q





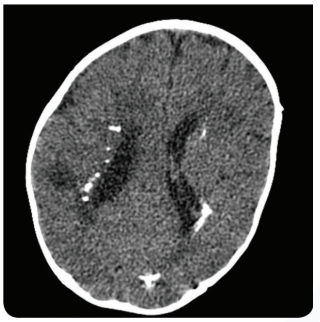
Pt. with fever + Convulsion and Altered Sensorium



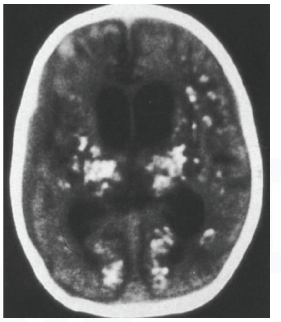


Intracranial Calcifications Q

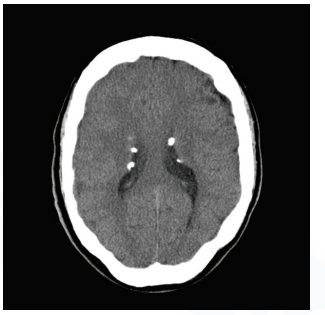
Periventricular



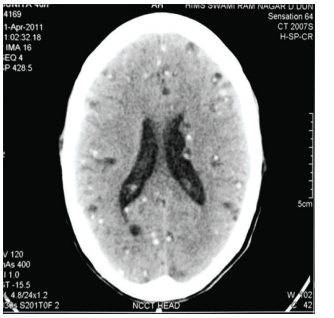
Diffuse Nodular

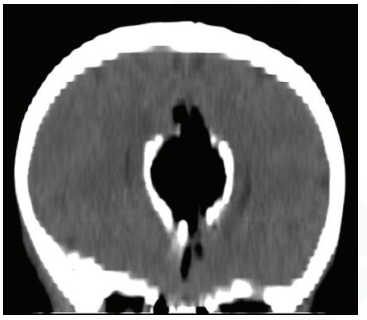


Subependymal

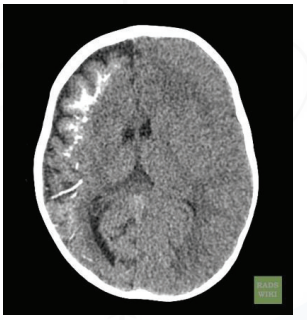


Starry sky





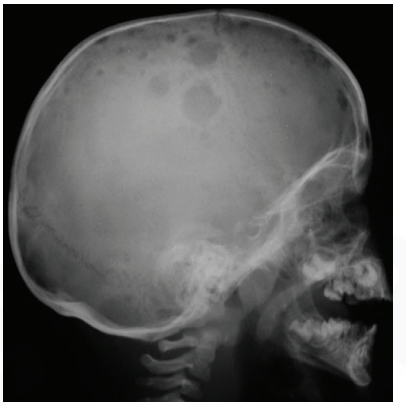
Bracket Calcification:
Corpus Callosum Lipoma



Tram Tract Calcification:
Sturge Weber Syndrome

Skull lesions and shape

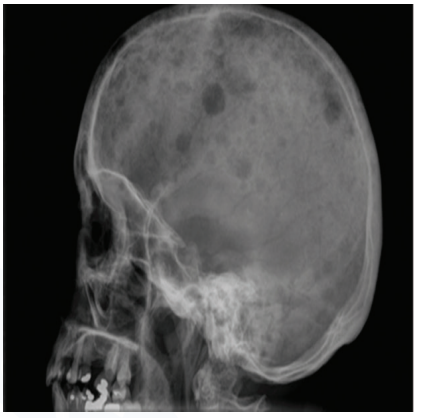
Hair on End:
Thalassemia / Hemo-lytic Anemia



Lytic lesion in Skull –

With bevelled Edges

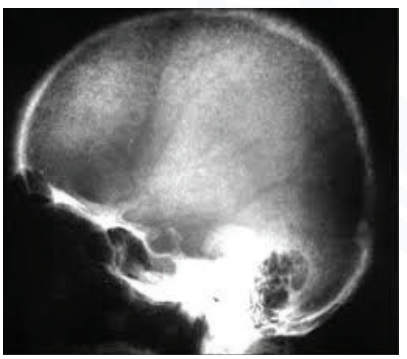
Eosinophilic Granuloma-Langerhans Cell Histiocytosis



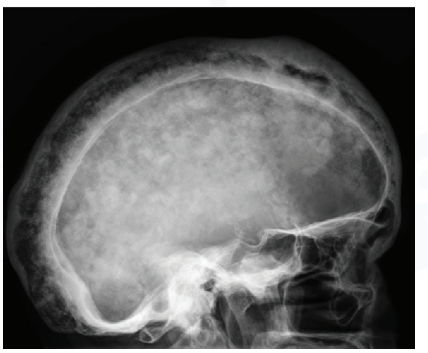
Lytic lesion in Skull –

Raindrop Skull / Cold Spot

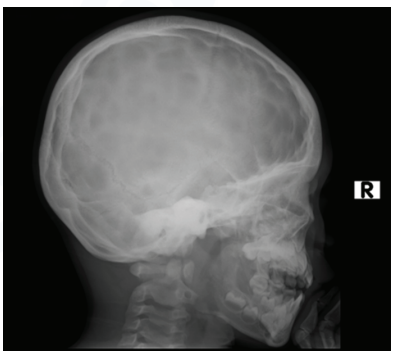
Multiple Myeloma



Pepper Pot / Salt and Pepper Skull
Hyperparathyroidism

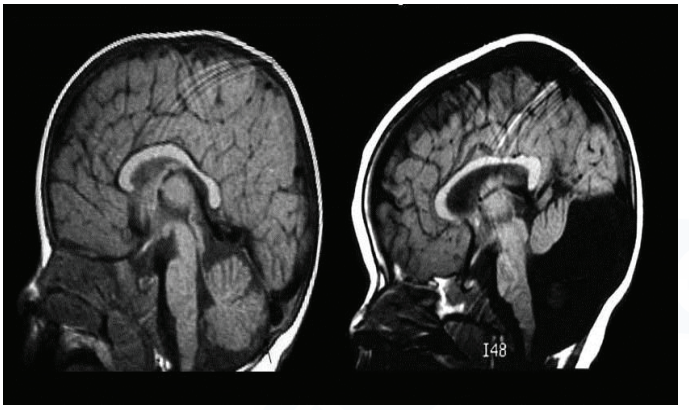


Cotton Wool Skull
Raised Alkaline Phosphate
Hot Spot on Bone Scan
Paget's Disease



Copper Beaten Skull
Indentation on the inner surface of the skull
Raised ICT- Hydrocephalus

Congenital Malformation : Brain Q



Chiari Malformation
Posterior Fossa

Tonsillar Herniation

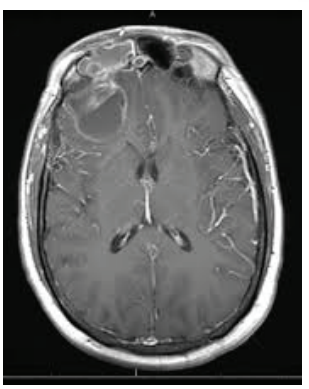
Dandy Walker Malformation

Cyst in Post. Fossa

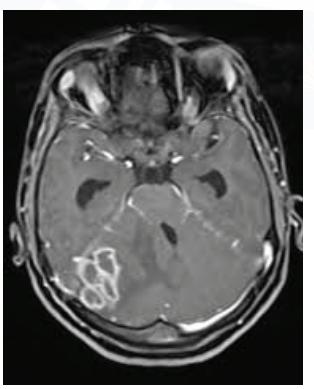
Chiari 1 + Myelomeningocele = Chiari II / Arnold Chiari Malformation

Q. Ring enhancing lesion (RENL) in following pts. Q

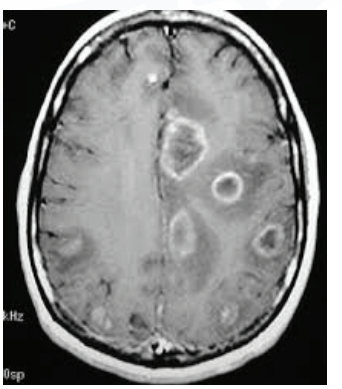
Sinusitis / Ear Discharge



Conglomerated RENL

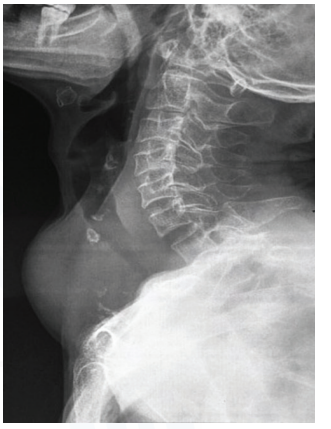


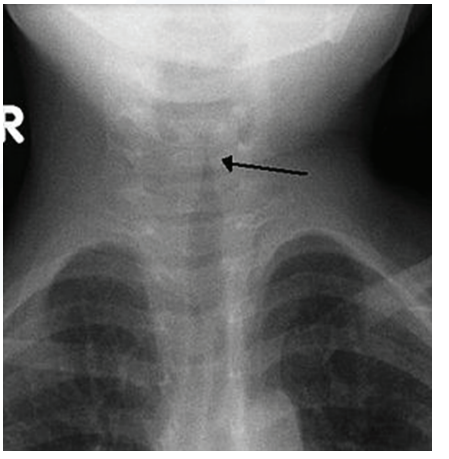
Immunocompromised: HIV / Organ Transplant



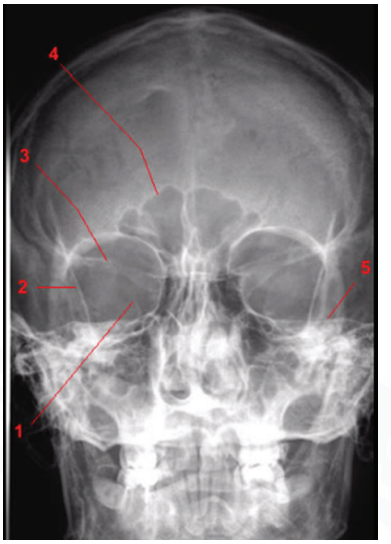
Child with seizure

HEAD AND NECK RADIOLOGY



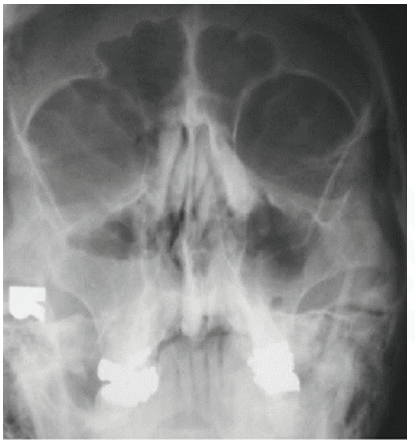


Waters View
Best for Maxillary Sinus
Pierre's- Open mouth water's view for sphenoid sinus[Q]
Posterior ethmoid cells NOT seen



Caldwell View[Q]
Best for Frontal Sinus
Ethmoid air cells seen
Also for Orbital rim

Lateral X-ray Skull
Best for Sphenoid Sinus
Also for Sella Turcica

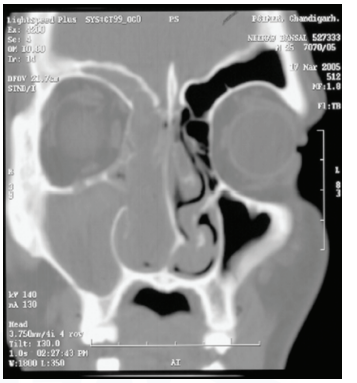


Air- Fluid Level in PNS
Acute Sinusitis
Compare with Orbit

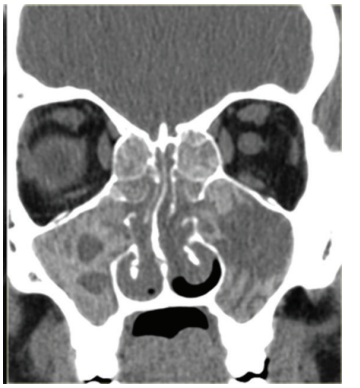
Normal NCCT PNS in Coronal Plane: Inv. Of choice for PNS disease

Abnormal NCCT PNS in Coronal Plane:
Mucosal Thickening

Identify the investigation:



Soft tissue in nose and Maxillary Sinus Unilaterally without Bony destruction -

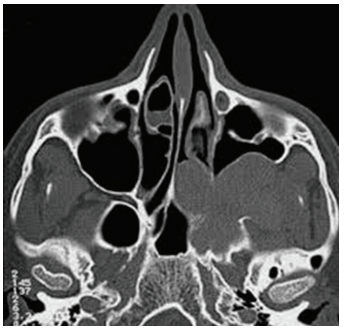


Soft tissue & Hyder density in nose and Maxillary Sinus Bilaterally without Bony destruction -

Q: Identify the sign shown below?

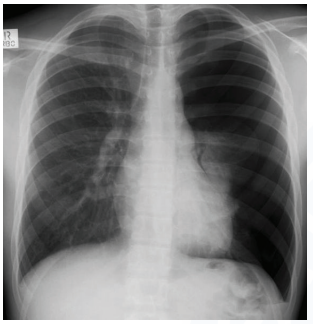
Q. Identify the sign shown in the below given image?

Q. In which pathology is it seen?

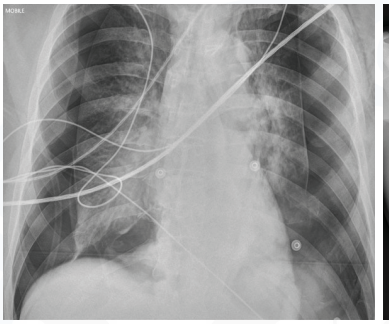


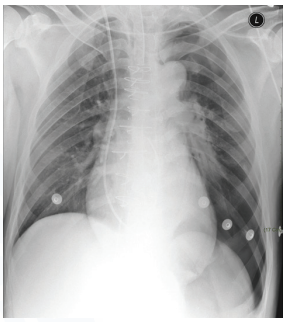
CHEST RADIOLOGY

Chest Imaging: Radio-lucency



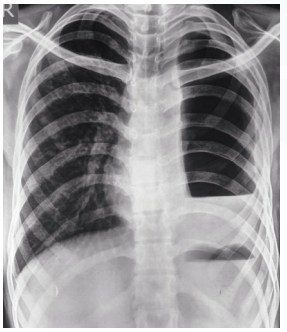
Lucency devoid of vascular markings...



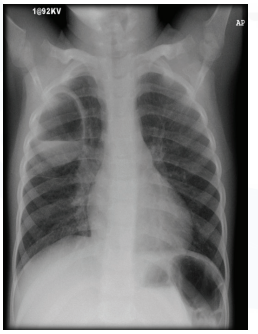


Sign of Pneumothorax on supine CXR

There are no straight lines on the radiograph, if you see one it represents Air-Fluid level[Q]

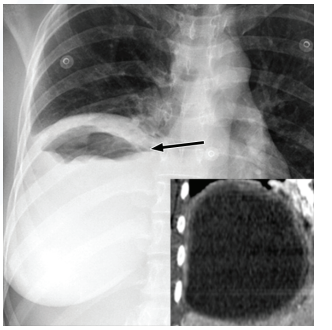


Physiological Air-Fluid level :
Air- Fluid level in Pleural Cavity:



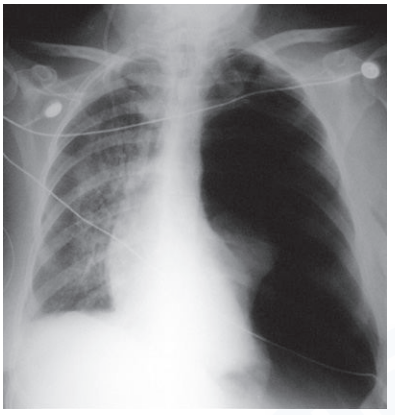
Air- Fluid Level in Lung

Air Crescent Sign:

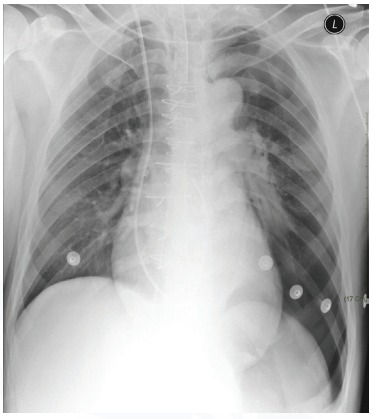


Lobulated/ Wavy Air-fluid level :

Sudden onset breathlessness Q



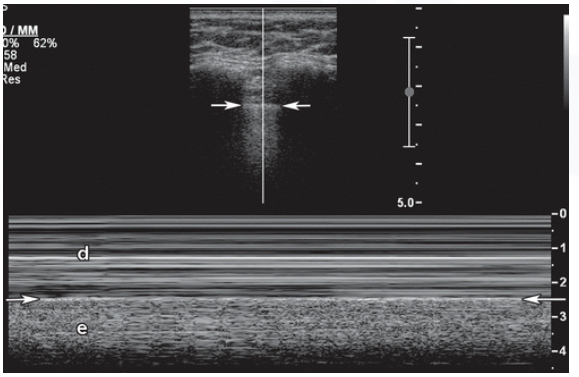
1. Clinical diagnosis
2. Collapsing Lung and Mediastinal shift indicates
3. Emergency Treatment



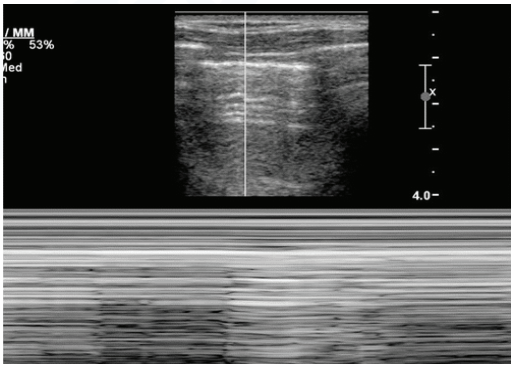
1. This sign is seen in Supine/ Erect Position
2. Sign of which condition
3. TWO findings include
   -
   -

Pneumothorax on M Mode USG Q

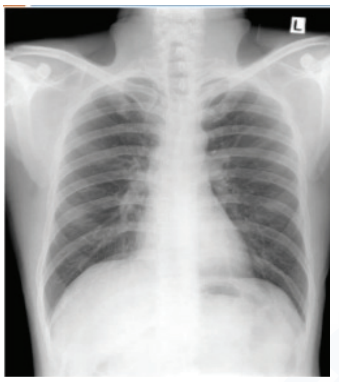
NORMAL LUNG MOVEMENT : PNEUMOTHORAX



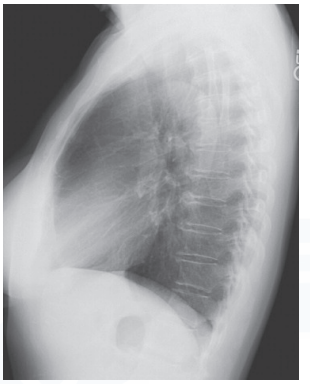
LUNG MOVEMENT NOT SEEN : NO PNEUMOTHORAX



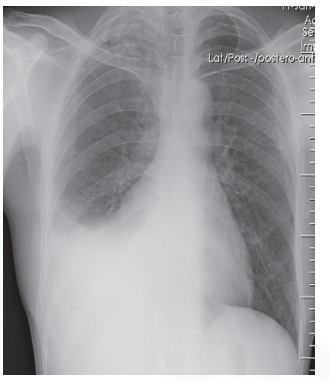
Costo-phrenic Angle:

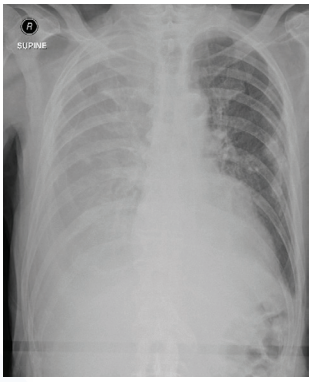


Most dependent part of the pleural cavity on Erect CXR .....

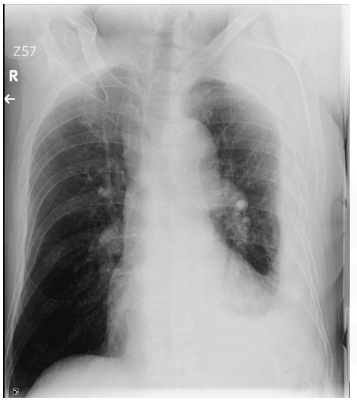


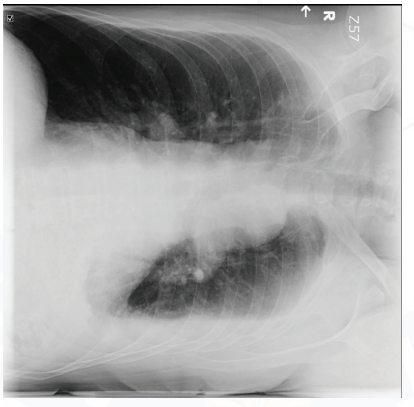
Minimal amount of fluid causing blunting of Lateral C-P angle.....

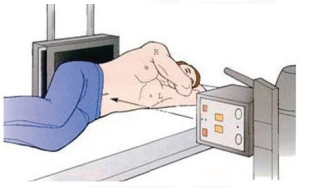




Minimal fluid on Supine CXR that can be detected.....





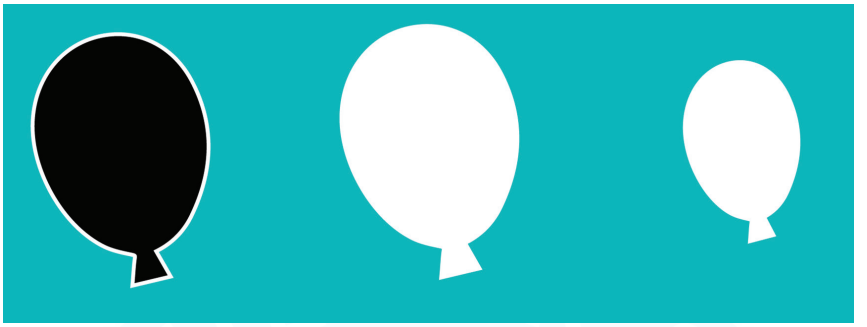


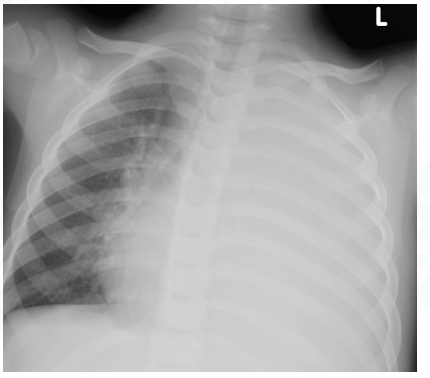
Most sensitive CXR for detection of Pleural effusion.....

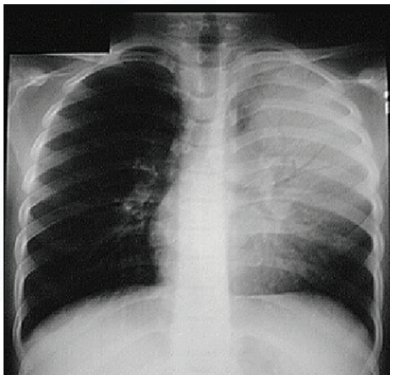
Minimal Amount of fluid that can be detected

Most sensitive radiological test for Minimal Pleural effusion

Radio-opacity / Opacity

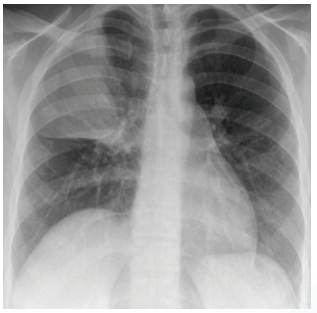
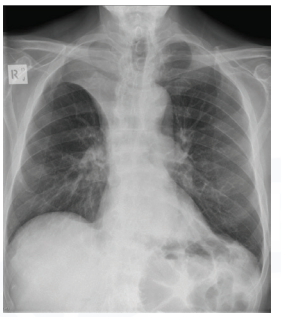






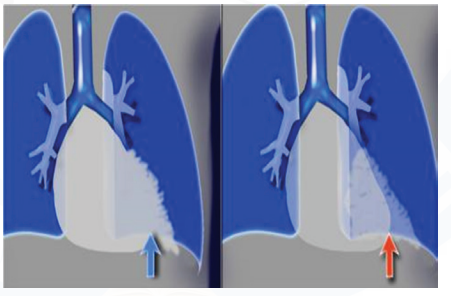
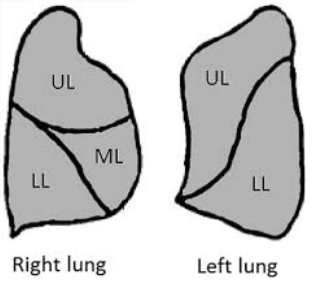
Morphology of Horizontal Fissure[Q]

Straight | Concave | Reverse S-shape | Bulging

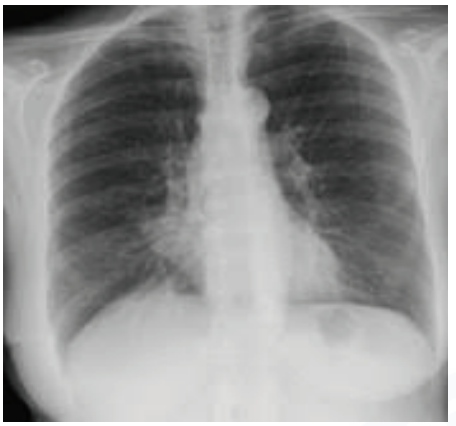
Lesion Localization

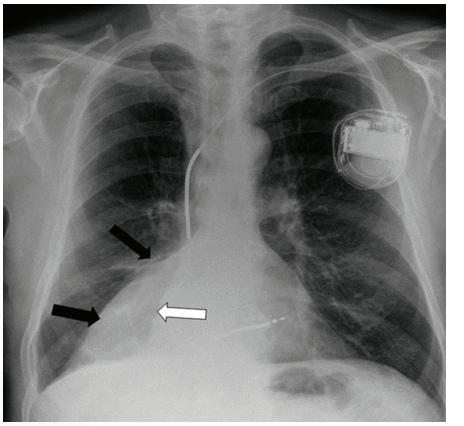
Silhouette Sign- If two structures are in contact then the touching border will not be seen.

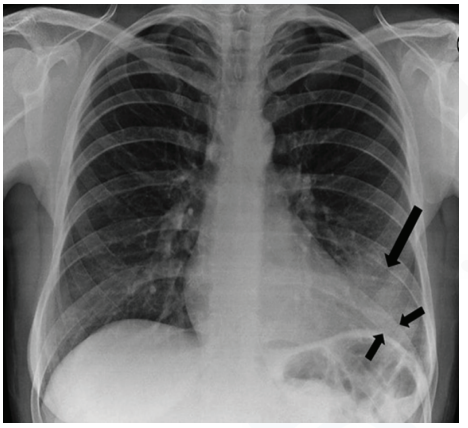



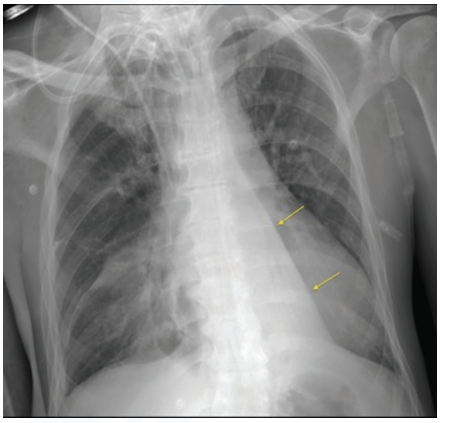
Right lung | Left lung

1. Heart is anterior structure in Mediastinum
2. Right middle lobe and Lingula are anterior
3. Both lower lobes are posterior



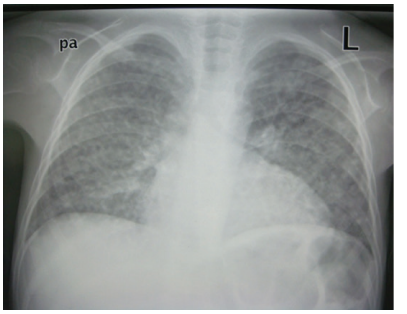


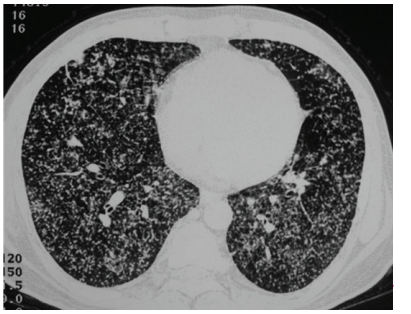




Q. Following show 'miliary shadows' on chest X-ray except:

1. Mitral stenosis
2. Tuberculosis
3. Staphylococcal pneumonia
4. Hemosiderosis

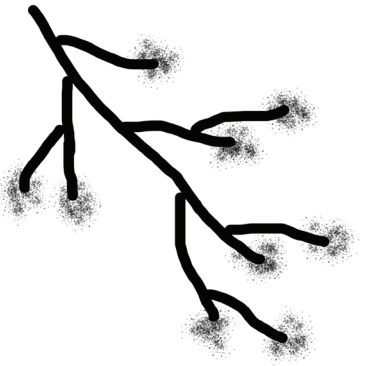


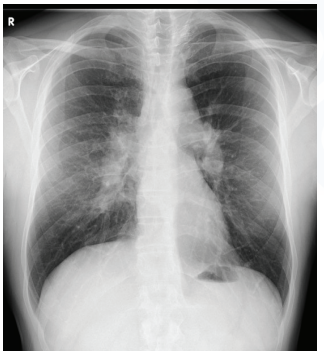


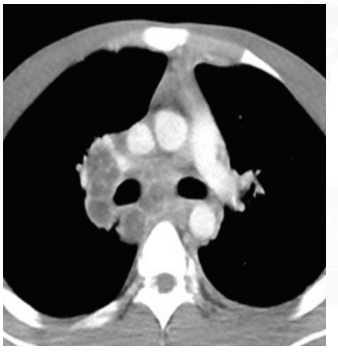
Blood Borne Spread

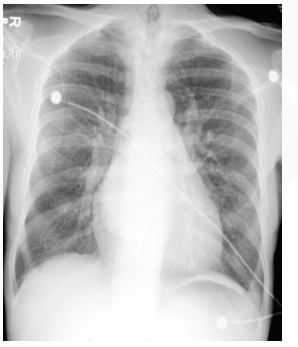
Spread through Airway

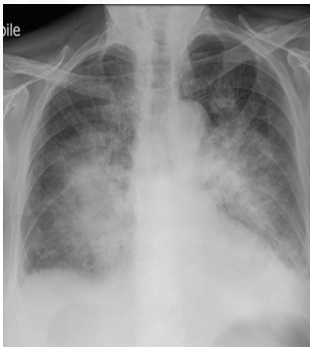
Reactivation/Reinfection :









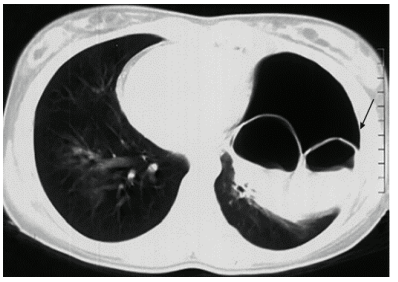


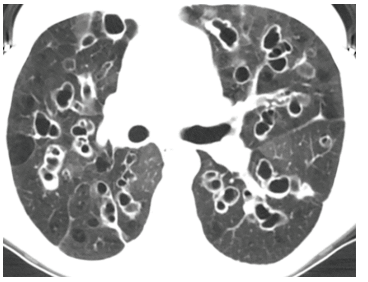
1. M/C cause of Bilateral Hilar Lymphadenopathy-
2. Conglomerated Necrotic Lymphadenopathy is seen in-
3. Other cause of Bilateral Hilar L.N. -

Sign of Early Pulm. Venous Hypertension

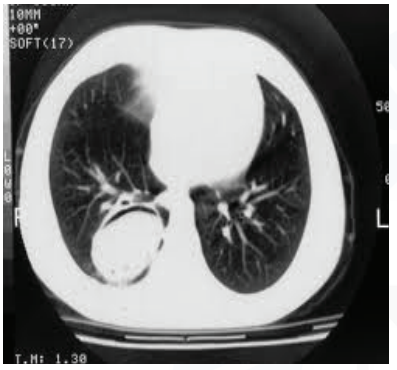
Sign of Advanced PVH

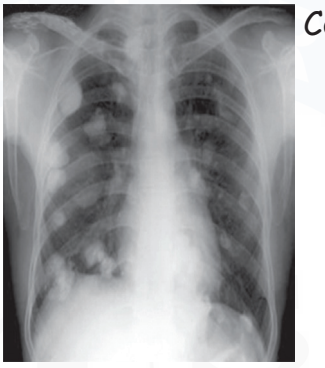
MIST





Associated with.....
Inv. Of Choice.....
Radiological Signs.....
1.
2.





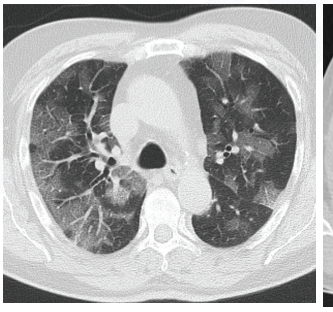
Commonly associated with

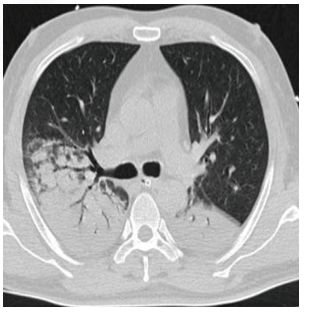
CORADS- Likelyhood of Infection

CT Severity Score -

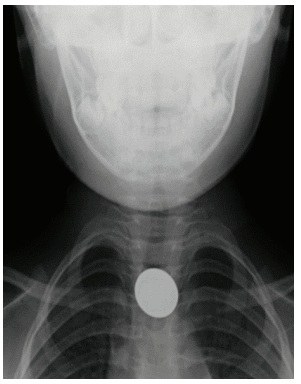
Classical Feature – Bilateral Lower Lobes Peripheral Ground Glass Opacities

Vessel seen or NOT through the Opacity



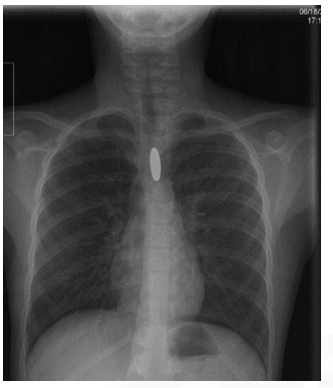


CO-RADS* Level of suspicion COVID-19 infection		
		CT findings
CO-RADS 1	No	normal or non-infectious abnormalities
CO-RADS 2	Low	abnormalities consistent with infections other than COVID-19
CO-RADS 3	Indeterminate	unclear whether COVID-19 is present
CO-RADS 4	High	abnormalities suspicious for COVID-19
CO-RADS 5	Very high	typical COVID-19
CO-RADS 6	PCR +	

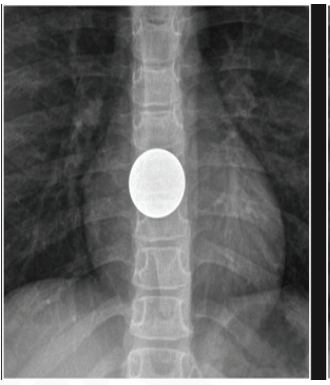


Face of the coin

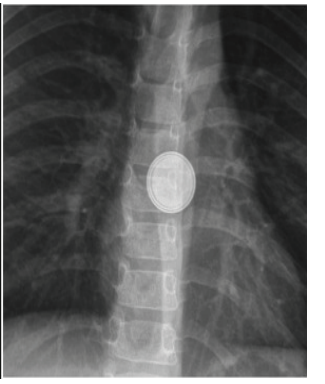
Frontal X-ray



Side of the Coin

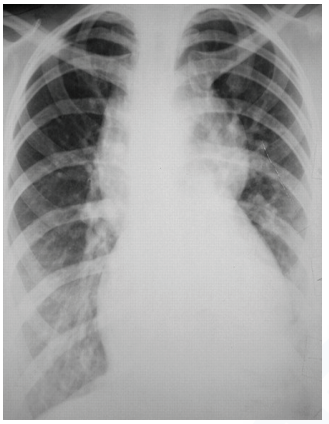


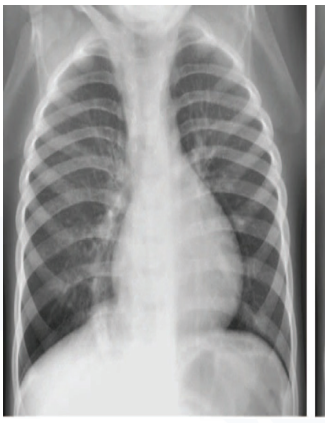
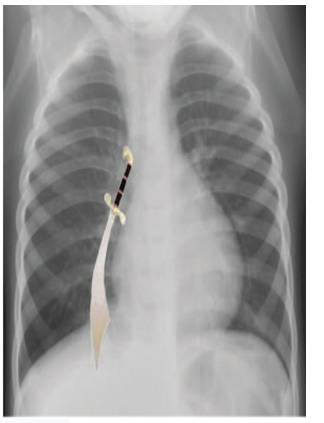
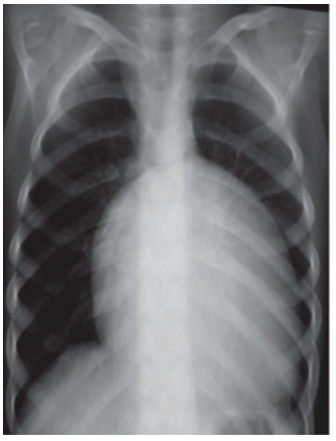
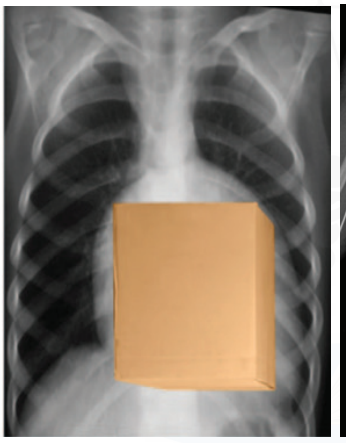
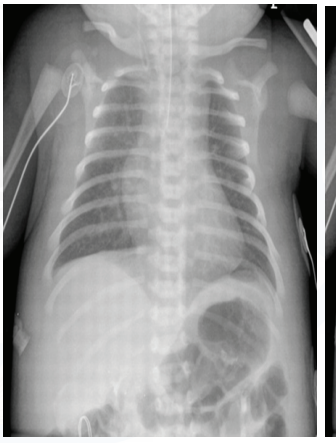
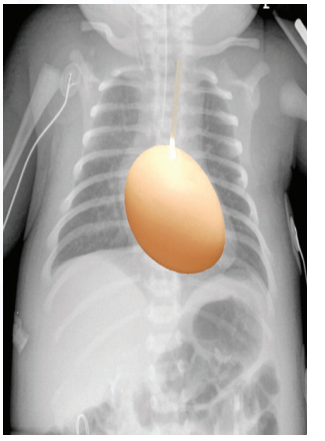
Homogenous

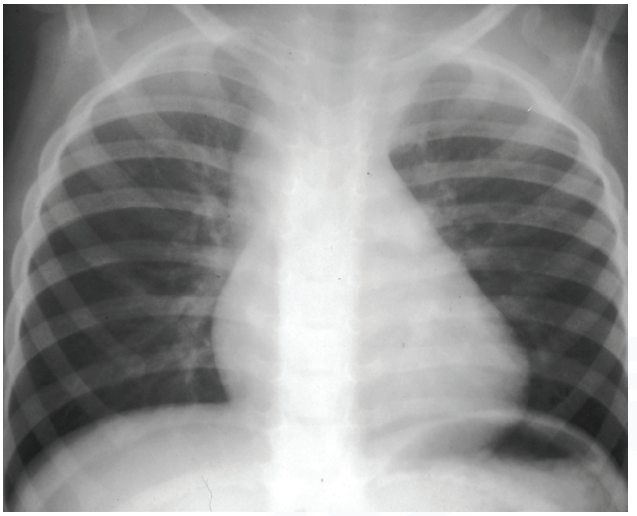


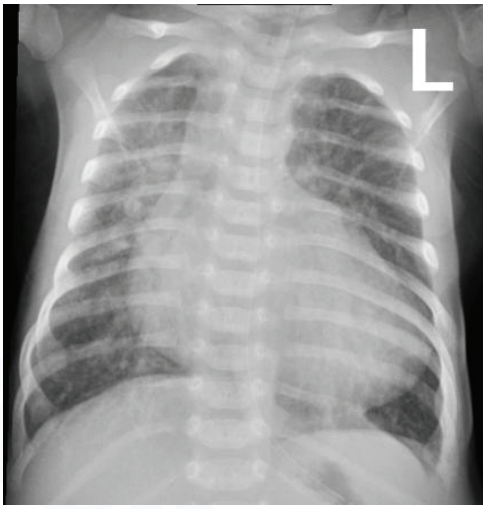
Double Ring

Cardiac Imaging



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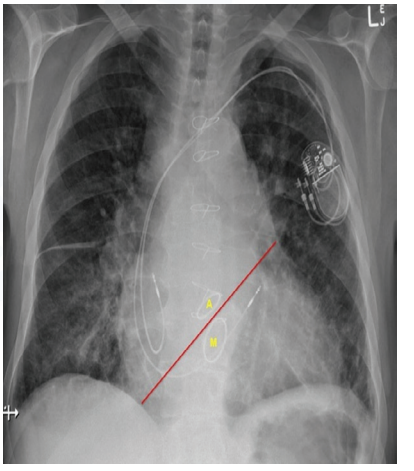
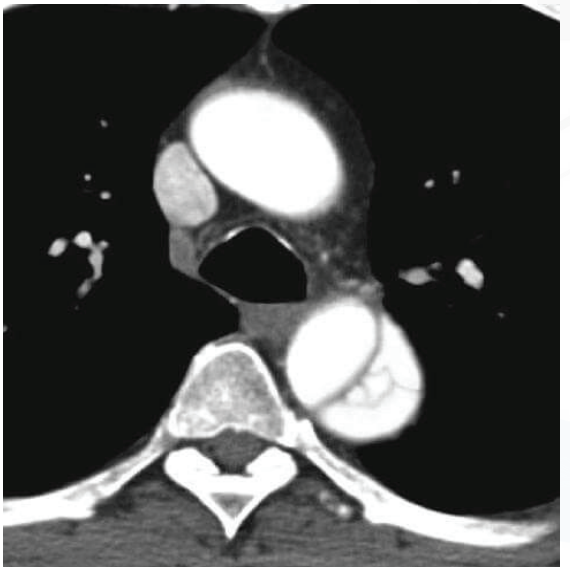
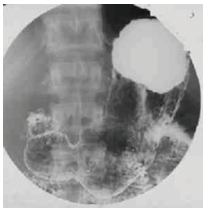




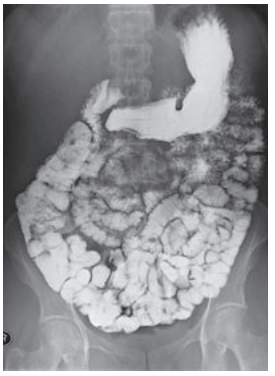
ABDOMINAL RADIOLOGY

Barium Sulfate Contrast Radiology

Types	Barium Swallow	Barium Meal	Barium Follow-through	Barium Enema
Procedure	Drink BaSO4 suspension	Ingest gas pellets and citric acid to expand the stomach. Then intake BaSO4.	Drink BaSO4 suspension. X-rays commonly taken at 0, 20, 40, 90 mins	Lubricated tube with BaSO4 inserted into rectum
Major site of use	Esophageal	Esophageal, stomach, duodenum	Small intestine	Large intestine
Limitations	●Risk of aspiration ●Poor mucosal detail ●Unable to biopsy	●Low sensitivity for early cancer ●Unable to biopsy or assess H. pylori	●Time-consuming ●Radiation exposure	●Difficult in frail elderly ●Uncomfortable ●Can miss polyps <1 cm ●Less useful for inflammations

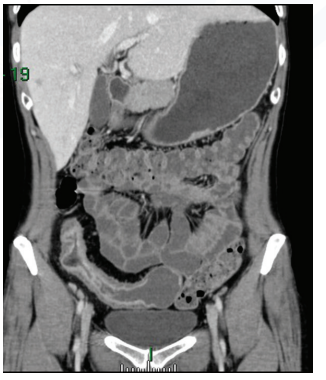


Barium Study of Small Bowel

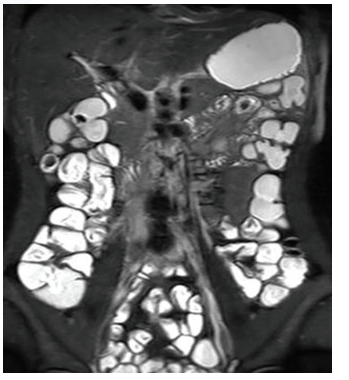


Barium study of the Small Bowel with Naso-duodenal Tube-

Inv. For Diagnosis of IBD Q



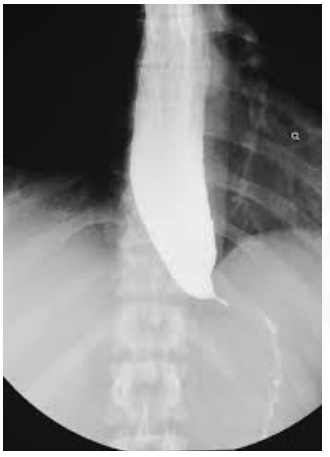
Inv. For Follow- UP of IBD Q



Oral Barium
Feathery Small Bowel
Procedure Name-

Per Rectal Barium Instillation
Incomplete Stricture Like
Large Bowel Procedure Name

Loss of Haustra
Ahaustral Large Bowel



Q. Shouldering sign in a pt. with dysphagia is a feature of which disease ?

1. Scleroderma
2. Malignancy
3. Achalasia Cardia
4. Diffuse esophageal spasm

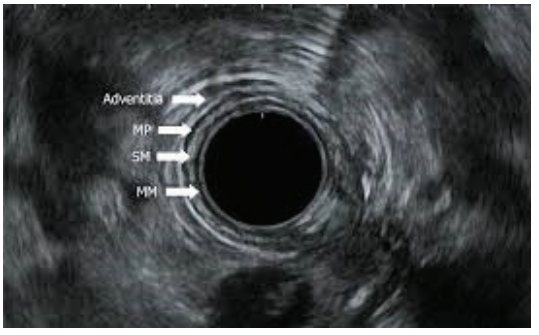
Q. 55 y/o female with non-progressive dysphagia especially to solids since 6 months. Barium meal study reveals irregular narrowing at distal end of esophagus with proximal dilatation. Diagnosis –

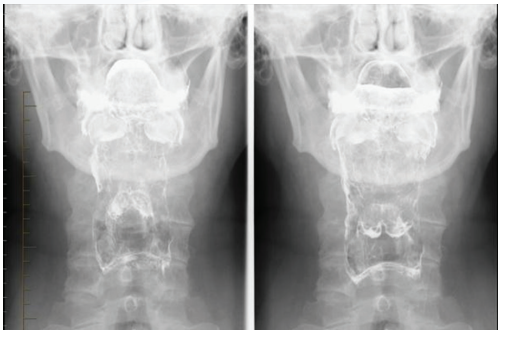
1. Achalasia Cardia
2. Peptic Stricture
3. Ca esophagus
4. Ca stomach involving the cardia

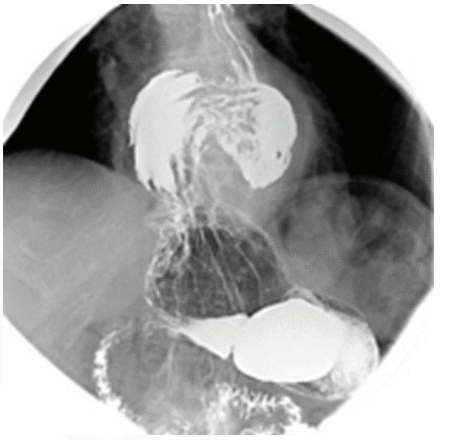
Q. Confirmatory test of choice for diagnosis of Achalasia Cardia?

1. Biopsy
2. Barium Meal Follow Through
3. Manometry
4. Endoscopic USG

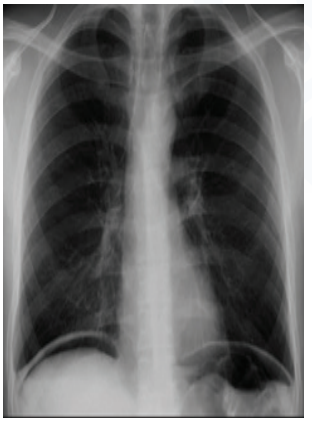
Q. Identify the investigation? Q

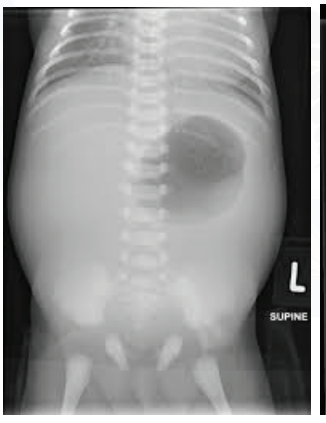


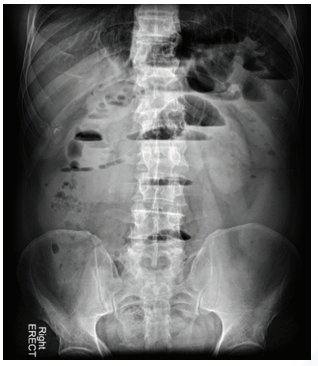




Patterns of Air-fluid level in Abdomen Q



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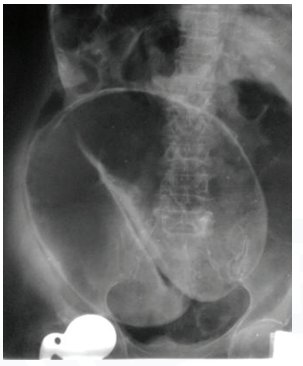

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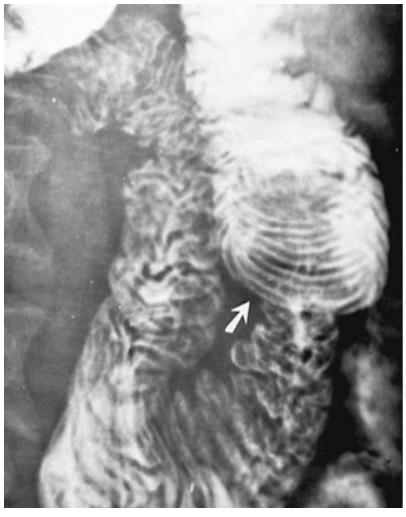
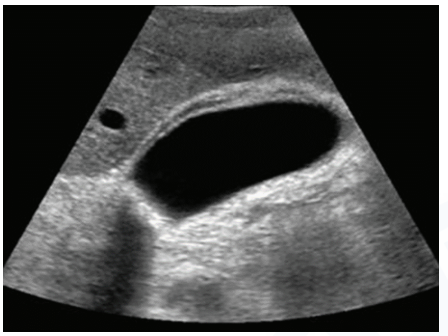
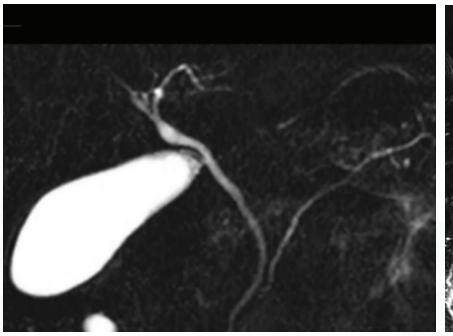
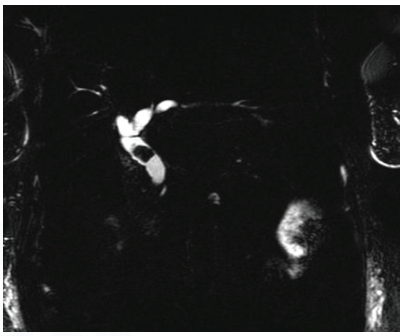
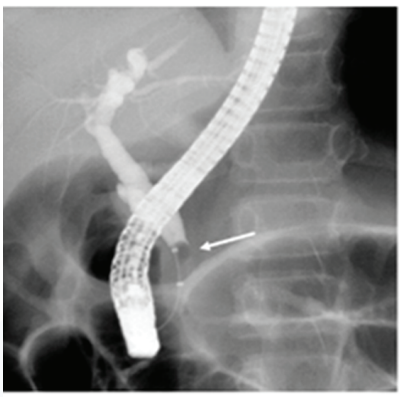
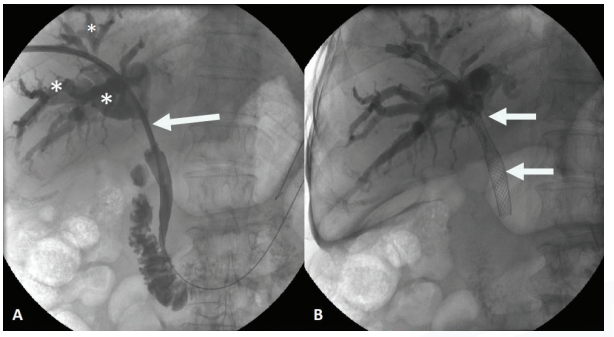
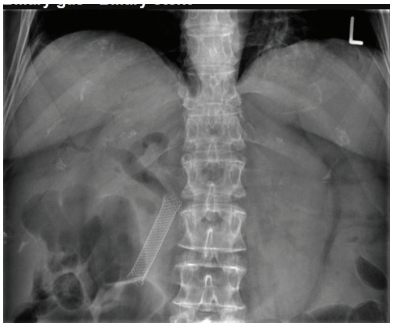






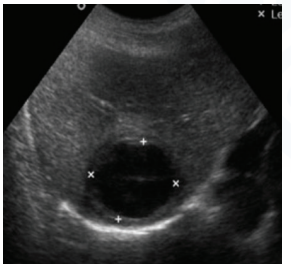


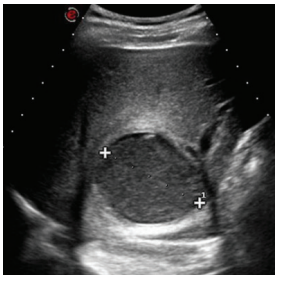
Two routes of Biliary Stenting

Two types of Stents are used:

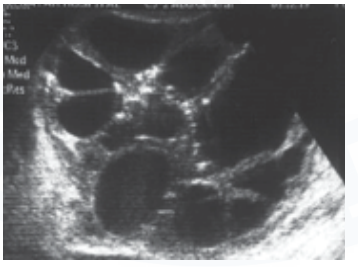
- Plastic stent
- Metallic stent (mesh like)

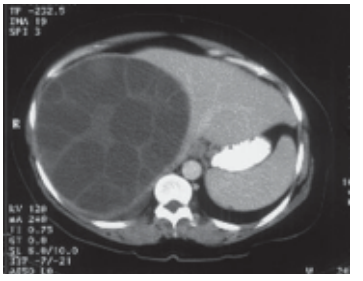
Cyst vs Abscess



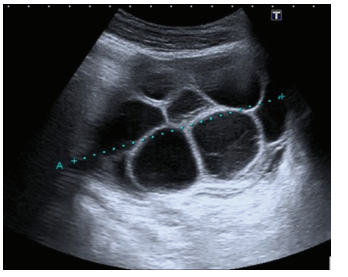


Multiloculated lesion in Liver

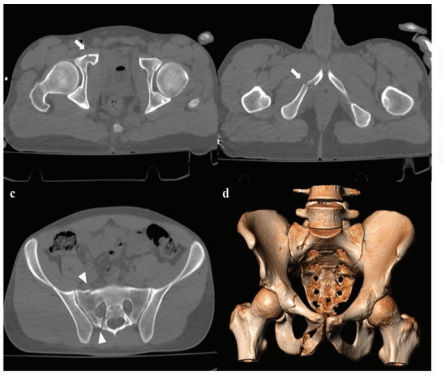




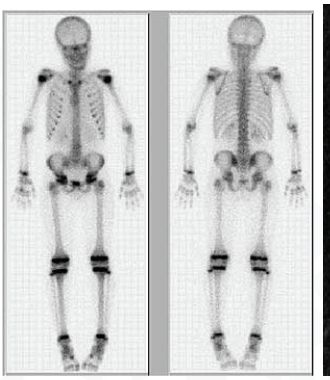
Multiloculated lesion in Ovary

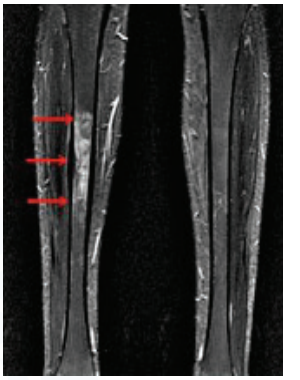


Fracture:
Simple



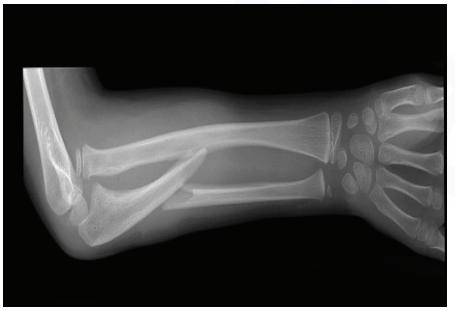
Complex Bone # - Skull / Spine / Pelvis Foot



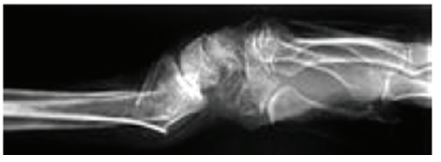


Fractures with Eponym

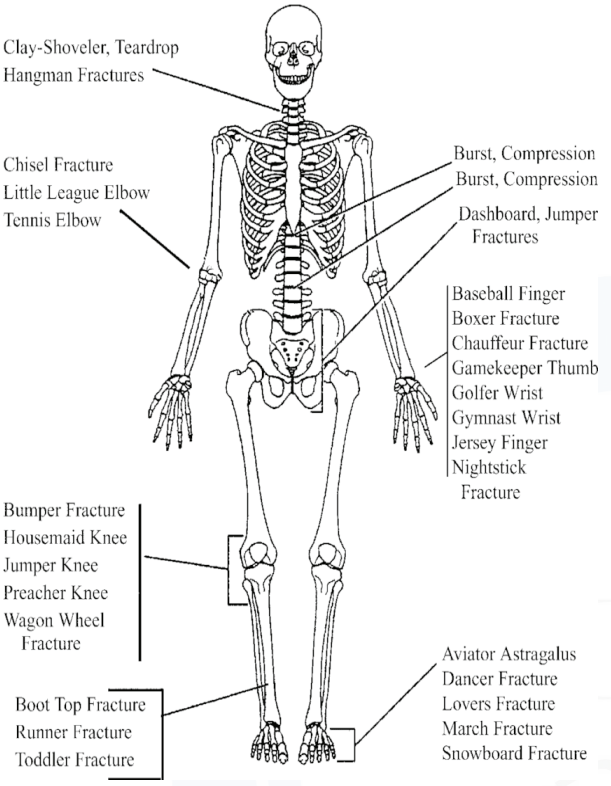
Monteggia's fracture-



Colle's Fracture



Galeazzi fracture



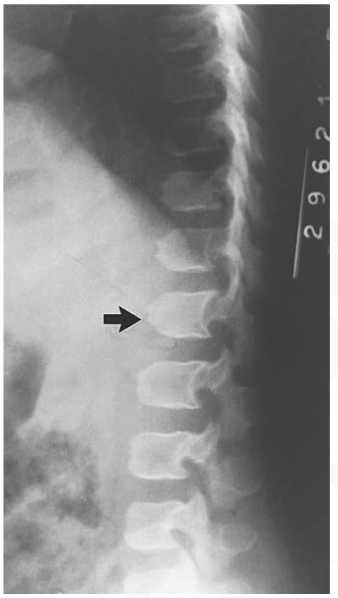
- C-I : Jeffersons #
- C-II : Hangman #
- C-7 /T-1 Spinous Process:

  Clay-Shoveler #

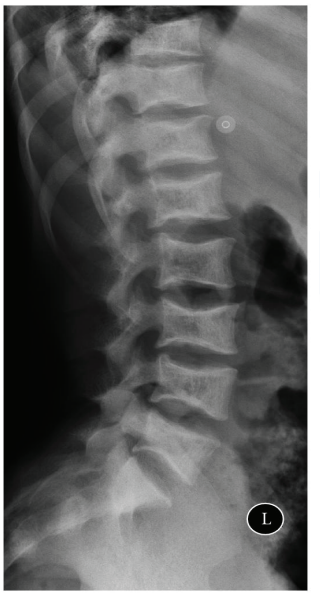
- CET – Lateral Epicondylitis – Tennis Elbow
- CFO – Medial Epicondylitis – Golfer's Elbow
- Apophysitis- Medial elbow pain in young pitchers – Little League Elbow
- Olecranon Bursitis – Student Elbow

Shapes Of Vertebrae Q

Bullet Shaped



Fish mouth



H-shaped Vertebra

Picture Frame Vertebrae

Ivory Vertebrae

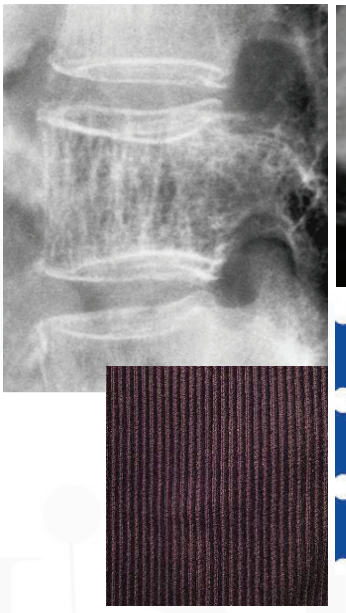
Vertebrae Plana

Potts Spine

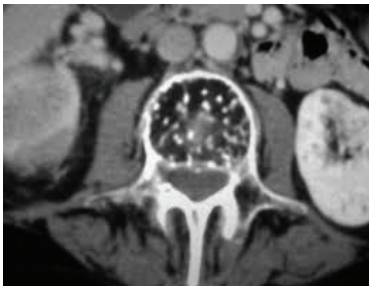
Sandwich vertebrae

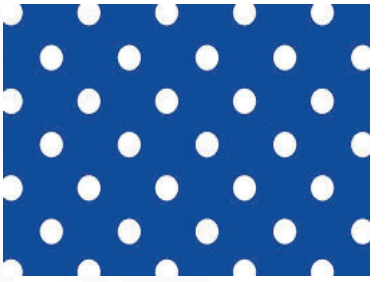
Rugger Jersey Spine

Corduroy

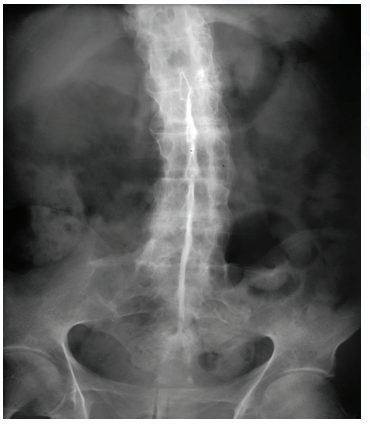


Polka Dot

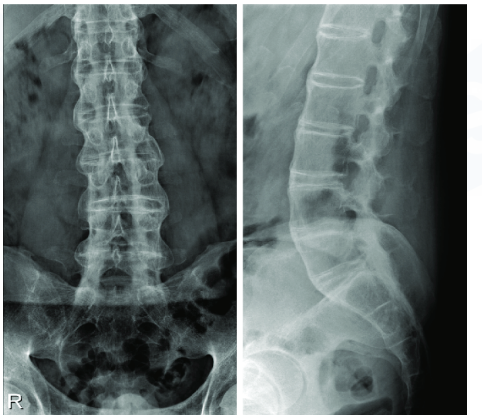




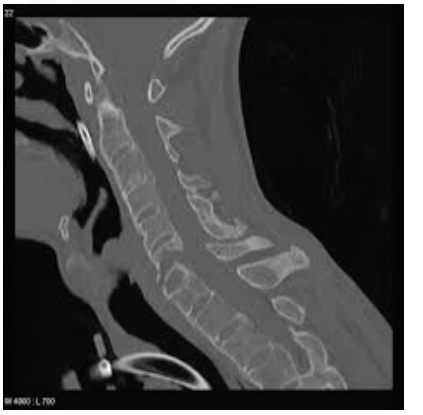
DAGGER SPINE



BAMBOO SPINE

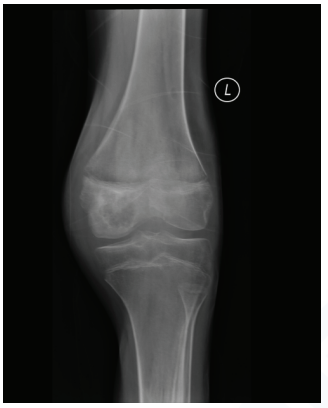


CARROT /CHALK STICK #



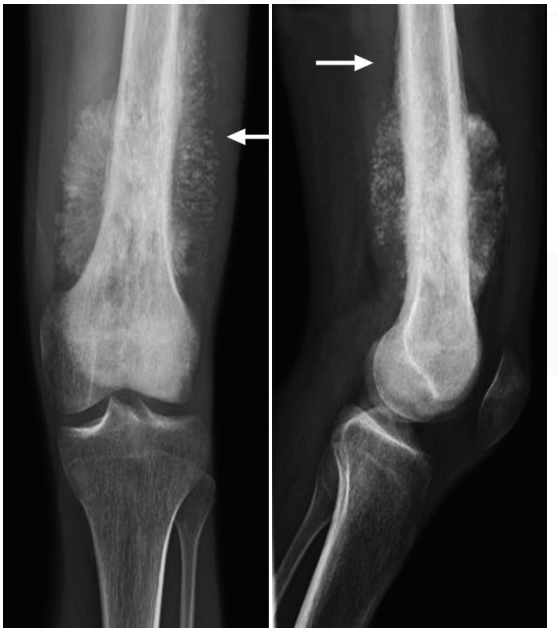
Bone tumour: Q

EPIPHYSEAL
GROWTH PLATE FUSED +/-



METAPHYSEAL
Within the confines of bone Outside

METAPHYSEAL MALIGNANT
TUMOUR



DIAPHYSEAL MALGNANT
TUMOUR :

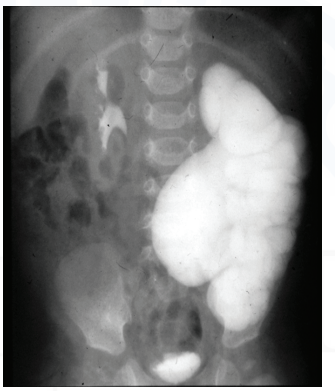
Coat Hanger Deformity

Epi plus Metaphysis Tumour

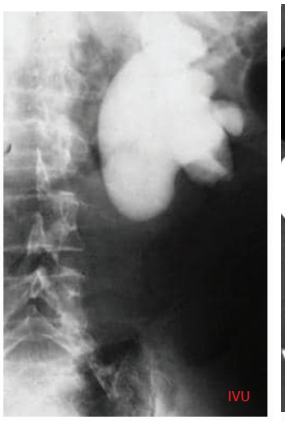
Shepherd Crook

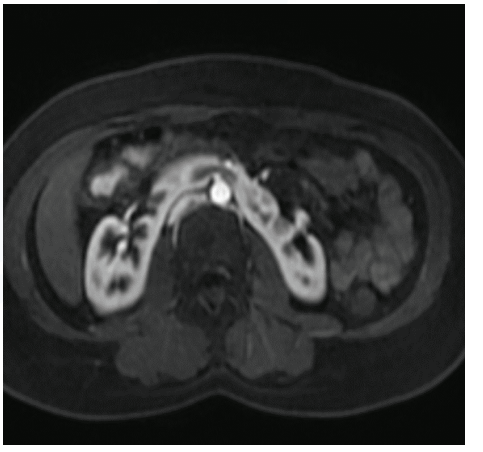
Genito-urinary System

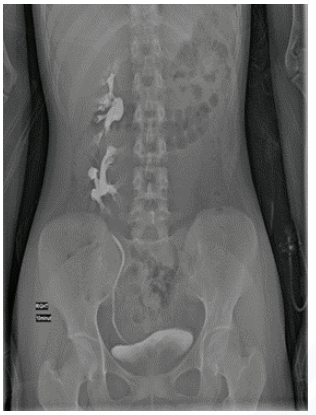
If contrast has been given then the white density is the contrast



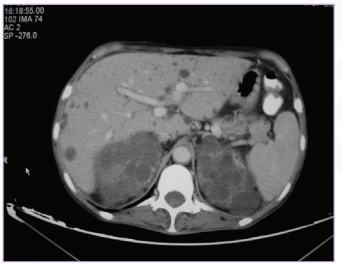
Question : Identify the investigation?Q

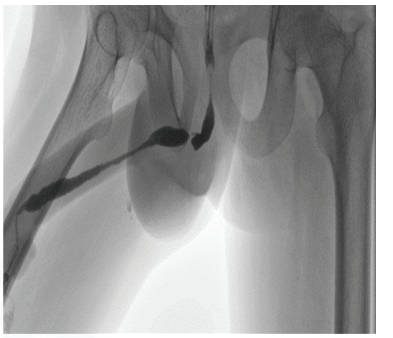


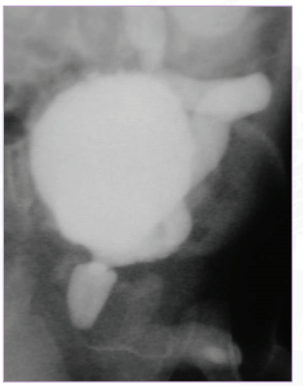


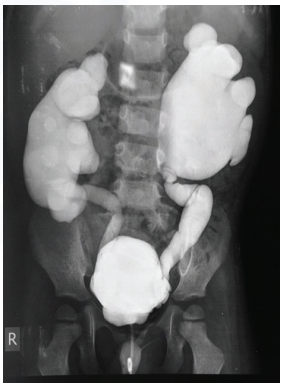


Right
15 MIN

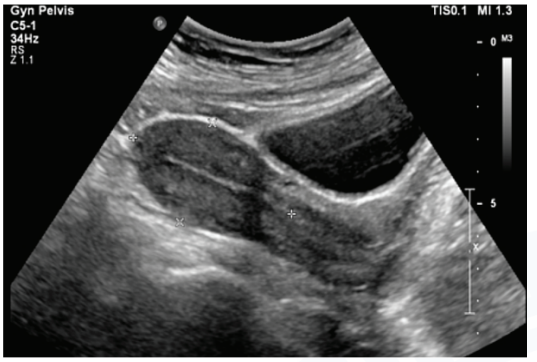

16:10:55.00
102 IMA 74
AC 2
SP -276.0

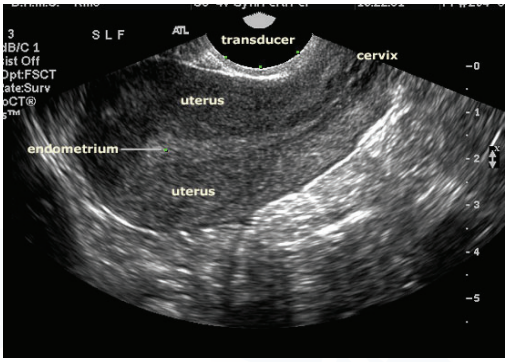


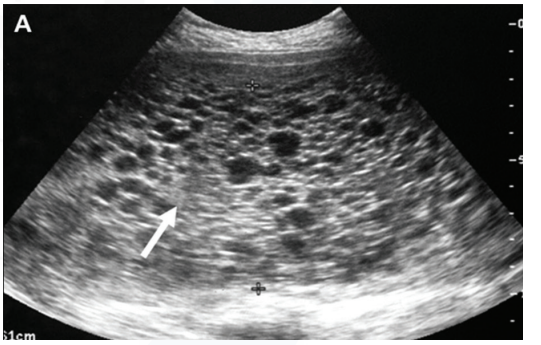


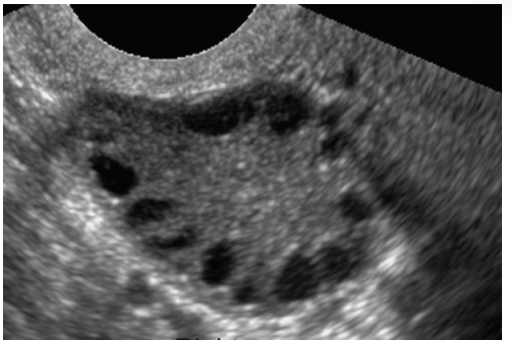


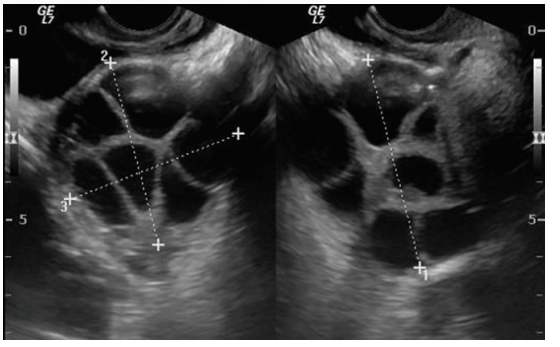
Pelvic USG


Gyn Pelvis
C5-1
34Hz
TIS0.1 MI 1.3

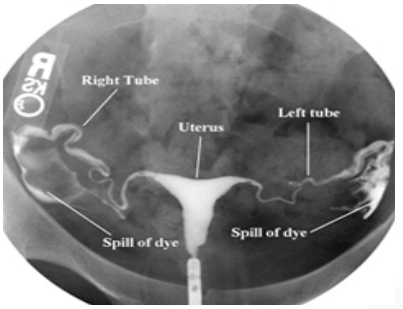

transducer
cervix
uterus
endometrium
uterus

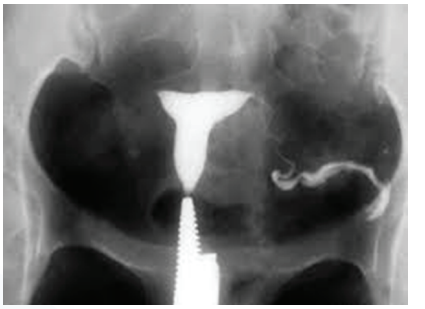

A

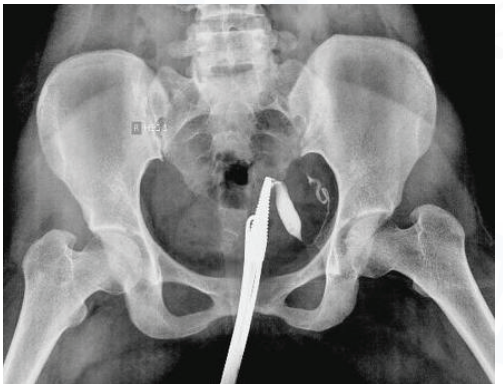




Hysterosalpingography

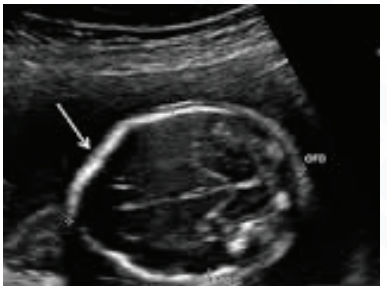






Obstetric USG : PC-PNDT Act

1. Dating Scan	Confirm the pregnancy Viability Twin gestation EDD
2. NT-NB Scan Screening for Aneuploidy	Nuchal Translucency Nasal Bone thickness
3. Anomaly Scan	Gross Congenital Anomaly
4. Growth Scan	Fetal Growth Assessment Doppler study



RADIATION PROTECTION AND RADIOTHERAPY

Radiation Protection

Lead Apron Thickness-

Personal Dosimeter TLD Badge- Made of Changed Every

Sign of

Sign of

Dose Limits (AERB)

Category	Dose Limit
Patient	50 mSv / year
Pregnant Patient	5 mSv/ term of pregnancy
Occupational worker	100 mSv over 5 year (but Annual equivalent is 30 mSv)
Pregnant Occupational Worker	2 mSv/ term of pregnancy
General Public	1mSv / year

Radiation Exposure in radiological Procedures

Diagnostic Procedure	Dose (mSv)
CXR (Single PA radiograph)	0.02
Mammography	0.2-0.8
IVU	2.5
CT Head	2.3
Bone Scan PET Scan Alone	4 5
Barium Enema	7
CT Chest	8
CT Abdomen	10
PET CT Combined	10-20

Radiotherapy: Radiation for treatment

Radiotherapy Types

Teletherapy

- Co60
- Cs137
- Linear accelerator

Brachytherapy

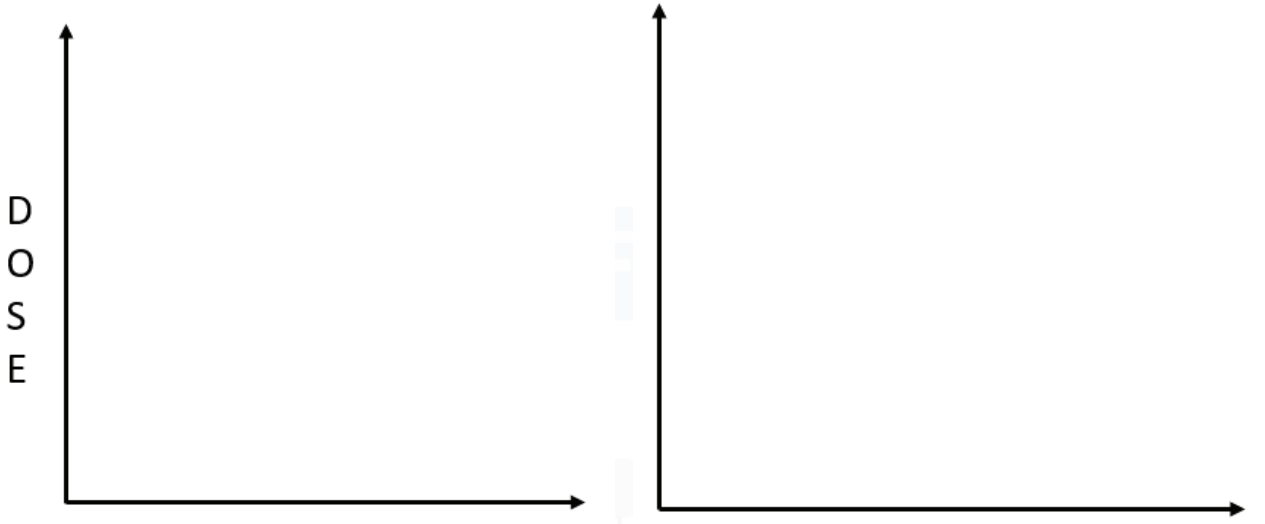
Interstitial - Intra cavitatory - Mould

Systemic Radionuclide

- I 131 – Thyroid cancer
- P 32 – Polycythemia Vera

Radiation in Teletherapy

Dose Depth Curve

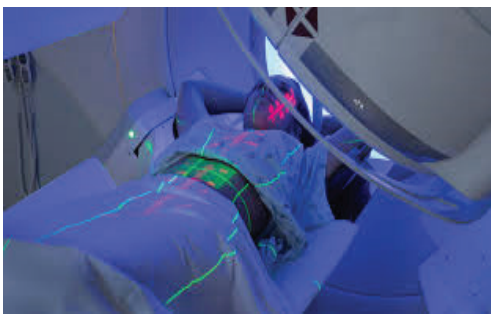


X-RAYS / GAMMA RAYS Electron Beam Proton Beam

Stereotactic Radiotherapy/Surgery

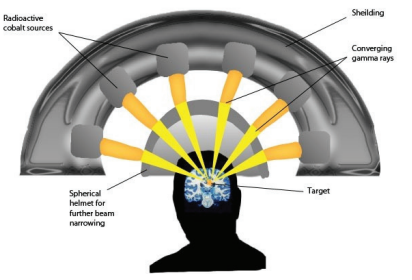
CYBERKNIFE

Radiation Used
Machine Used-
Helmet-
Robotic Arm
Indication

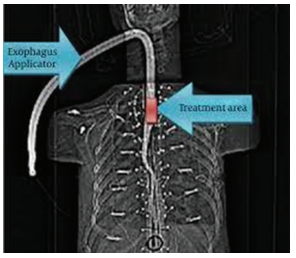


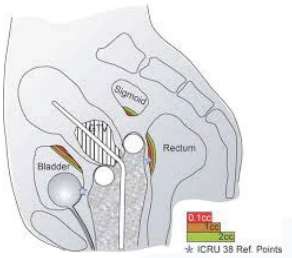
GAMMA KNIFE

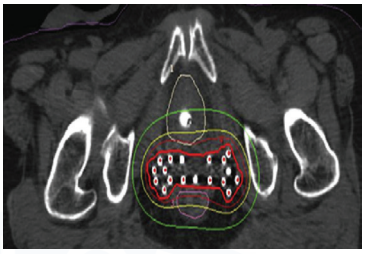
Radiation Used
Machine Used-
Helmet- Leksell Frame
Robotic Arm
Indication

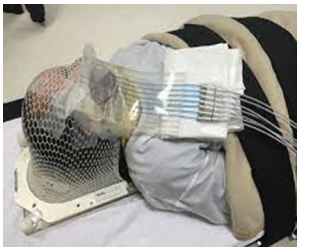


Intra Luminal -Intra cavitatory-Interstitial-Mould BT[Q]









Radio-isotopes[Q]

Emission of Beta Rays by	Both Beta+ gamma rays
Ytterium	Gold
Phosphorus	I- 131
Strontium	Radium

isotope	Half life
Tc99	6 hours
I123	13 hours
I125	60 days
I131	8 days
I132	2.3 hours
P32	14 days
Co60	5.2 years
Ra 226	1622 years

Radiosensitivity Q

	Most	Least
Stage of cell cycle	G2M	S
Organ	Ovary >>testis	Vagina>bone>cns
Tissue	bone marrow	Nervous tissue
Cellt type	undifferentiated, well nourished, divide quickly and are highly metabolically active	quiscent
Blood cell Eye	Lymphocyte Lens	Platelet Sclera

Highly sensitive	Least Radiosensitive
Wilms	Hepatoma
Ewings	Osteosarcoma
Lymphoma	Melanoma
myeloma	Pancreatic Carcinoma
seminoma	
WELMS	**HOMP**